# Large Language Models (LLMs)

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# Large Language Models

BCV

## Application Layer

**Copywriting**
Jasper
copy.ai
Headline
co:here
HyperWrite
Writesonic
Contenda
unbounce
copysmith

**Coding**
tabnine
MUTABLE AI
Codiga
co:here
GitHub Copilot
CODEGEN

**Dev Tools**
algolia
warp
Mintlify
cogram
Debuild
repl.it

**Chat / Comms**
MessageBird
Replier.ai
Sapling
FABLE

**BizOps**
viable
Enterpret
tabulate
Anecdote
OTHERSIDE AI
casetext
Dover

## Infrastructure Layer

**Model Creation**
AI21 labs
OpenAI
NVIDIA
Adept
EleutherAI
ANTHROP\C
Google AI

**Hardware**
habana
NVIDIA
SambaNova SYSTEMS
cerebras
GRAPHCORE

**Fine Tuning**
Google AI
aws
OpenAI
Hugging Face

**Inference**
OpenAI
Hugging Face

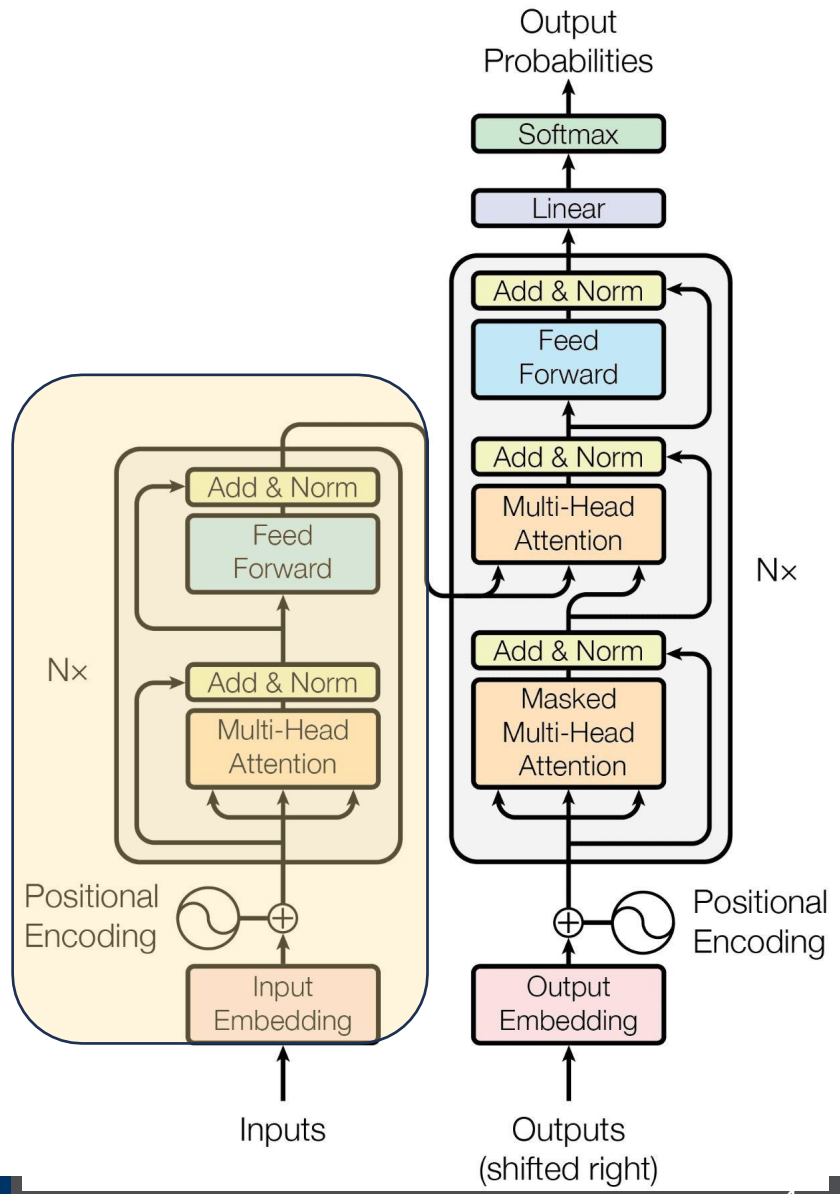
# BERT and GPT Concepts

- **BERT (Bidirectional Encoder Representations from Transformers)**
  - Uses transformer encoder only.
  - Trained with Masked Language Modeling (MLM).
  - **Bidirectional:** Considers both left and right context.
  - Fine-tuned for tasks like QA, classification.
  - **Architecture:**
    - Layers of encoder blocks.
    - Positional encodings added to embeddings.
    - Self-attention heads capture dependencies.

# BERT - Bidirectional Encoder Representations

- One of the biggest challenges in LM-building used to be the lack of task-specific training data.

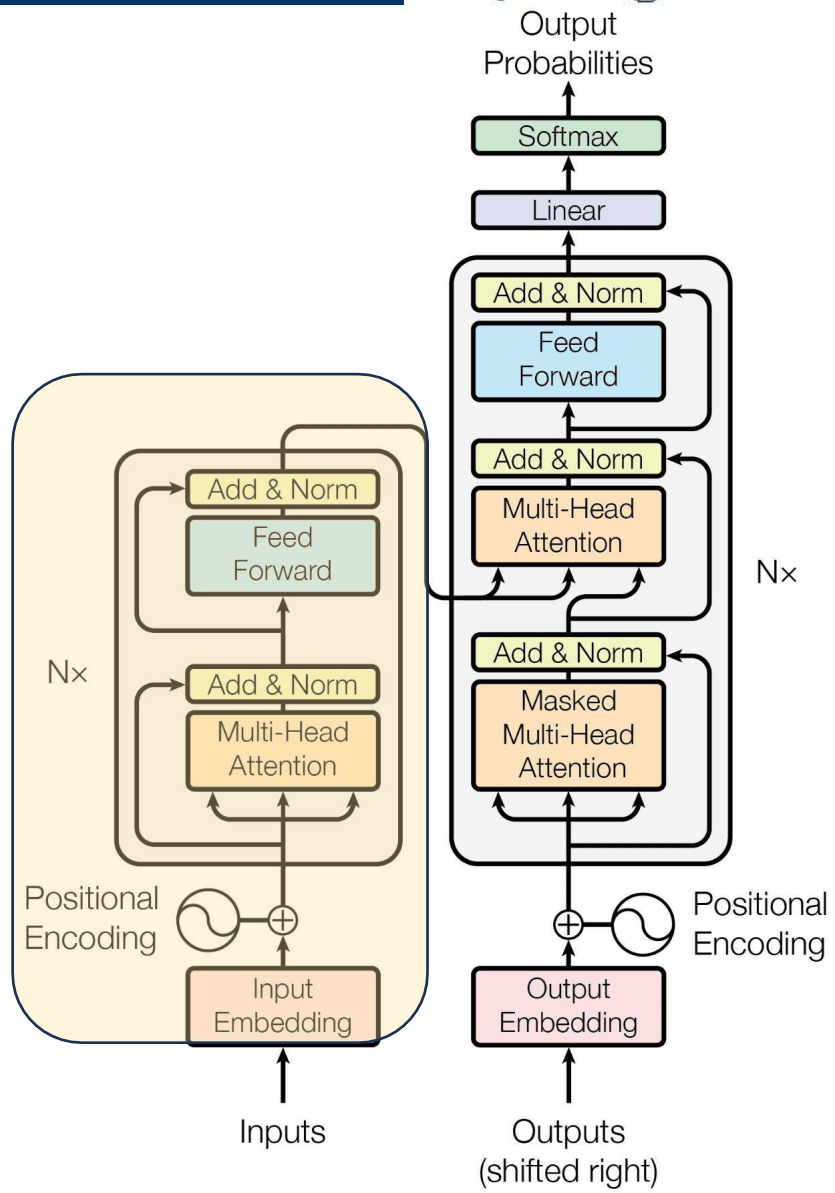
- What if we learn an effective representation that can be applied to a variety of downstream tasks?
  - Word2vec (2013)
  - GloVe (2014)



# BERT - Bidirectional Encoder Representations

**BERT Pre-Training Corpus:**

- English Wikipedia - 2,500 million words
- Book Corpus - 800 million words



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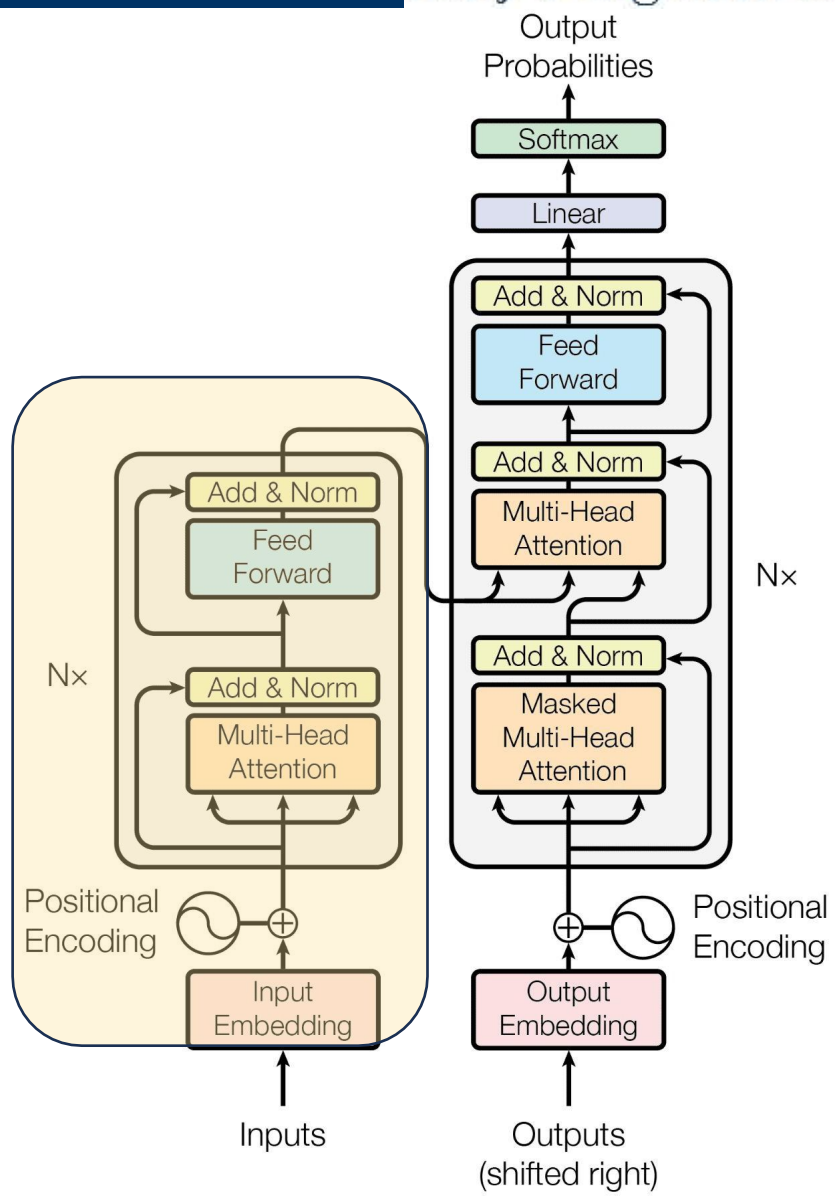
- MLM (Masked Language Modeling)
- NSP (Next Sentence Prediction)



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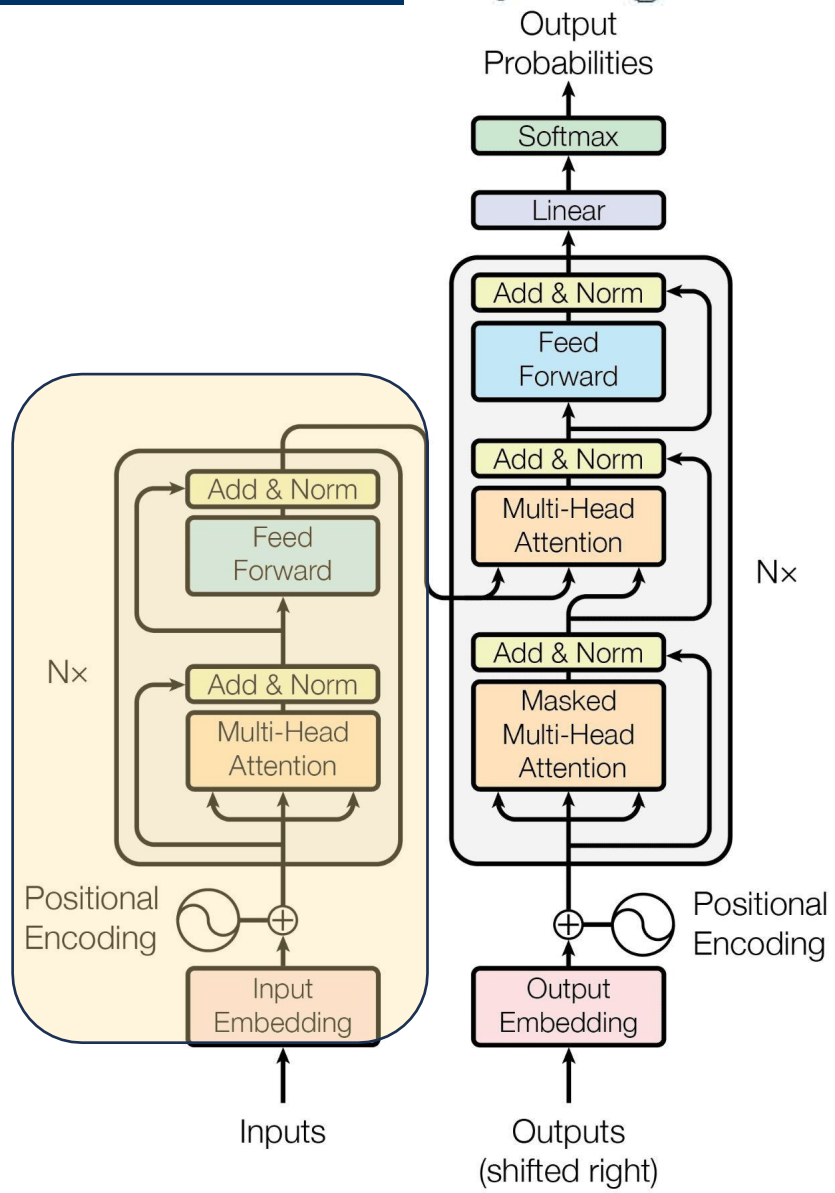
**BERT Pre-Training Results:**

- BERT-Base – 110M Params
- BERT-Large – 340M Params



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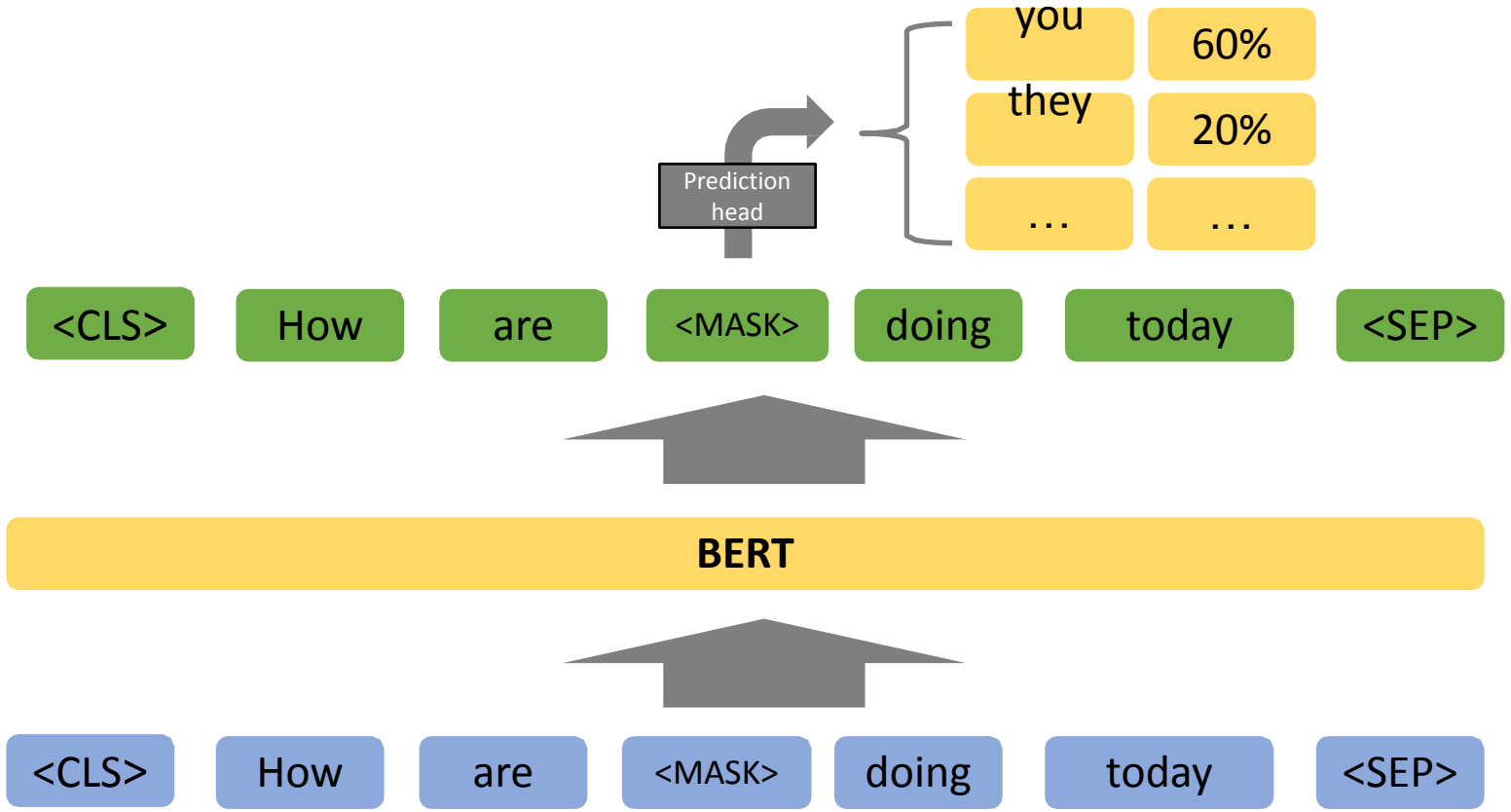


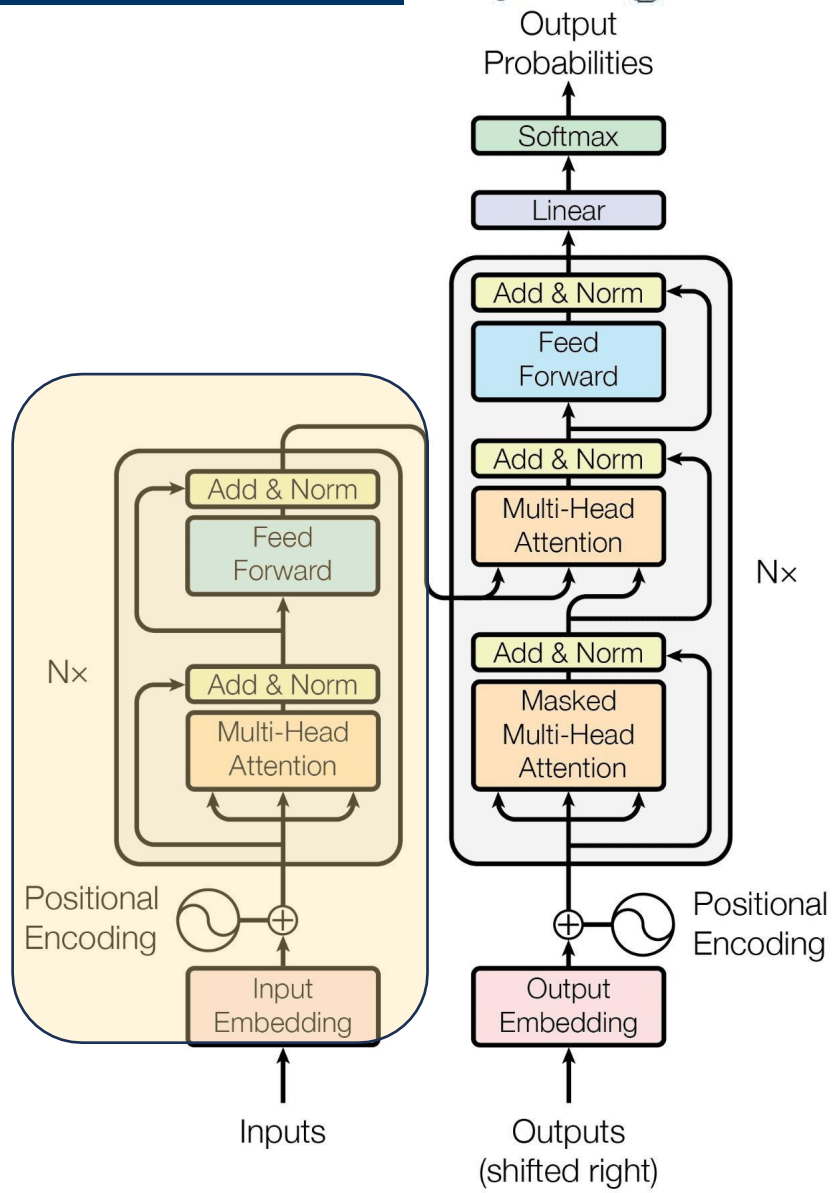

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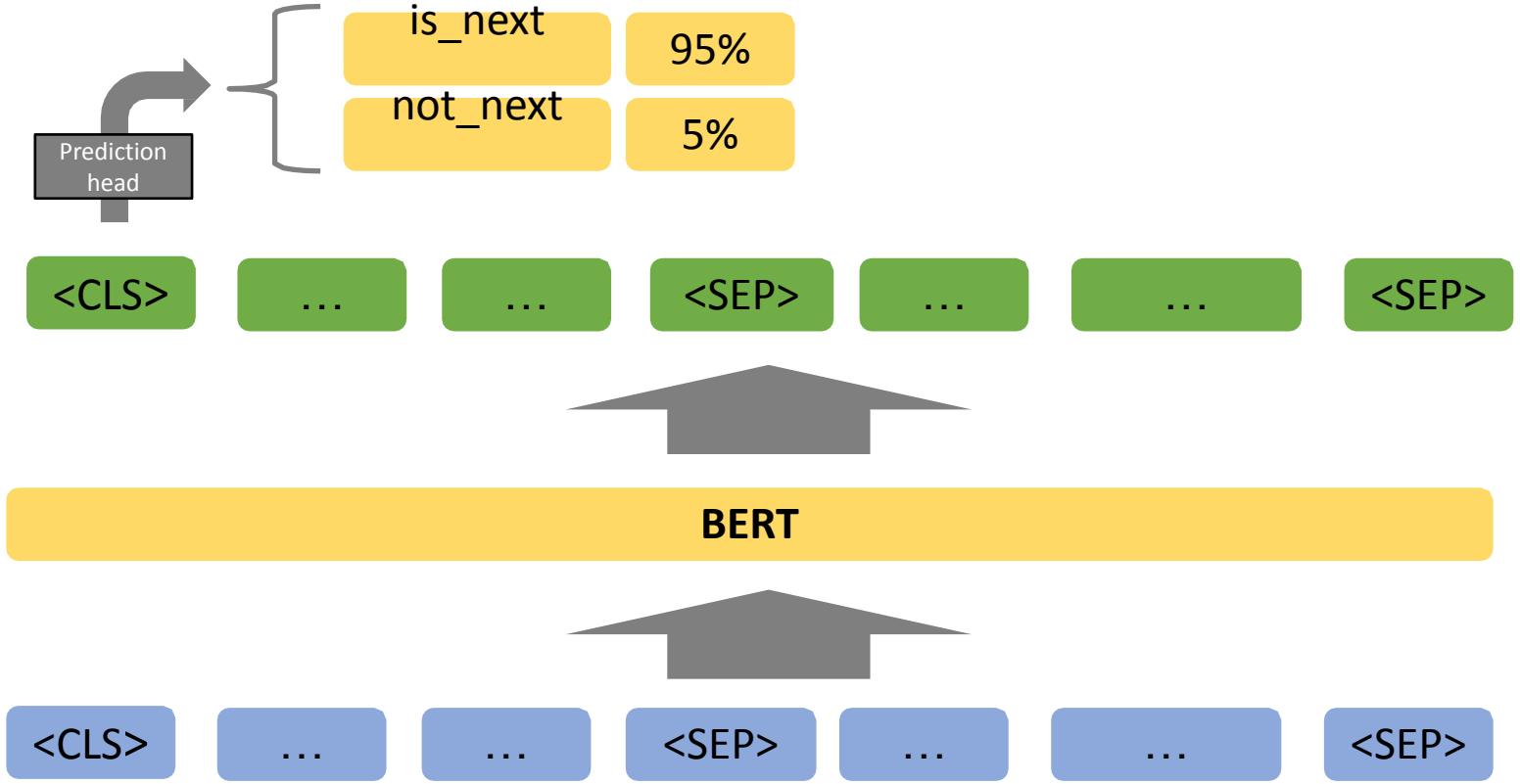
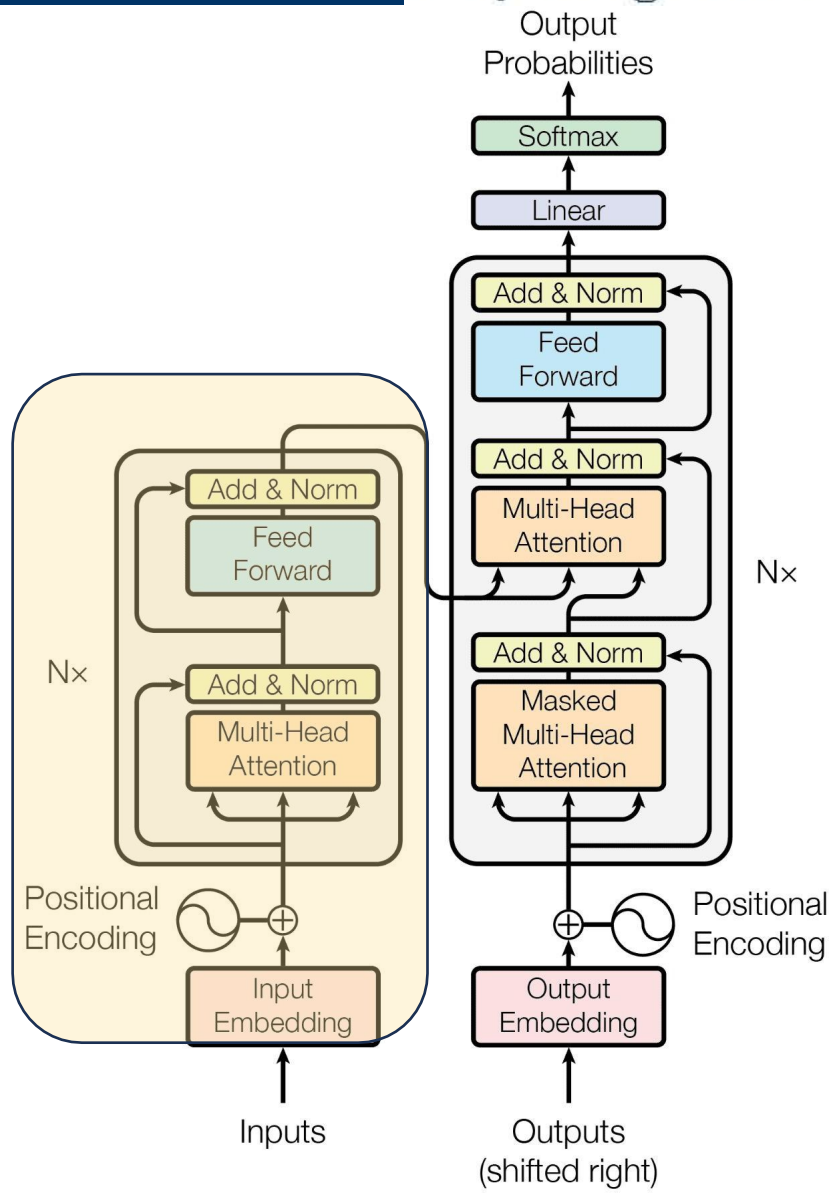
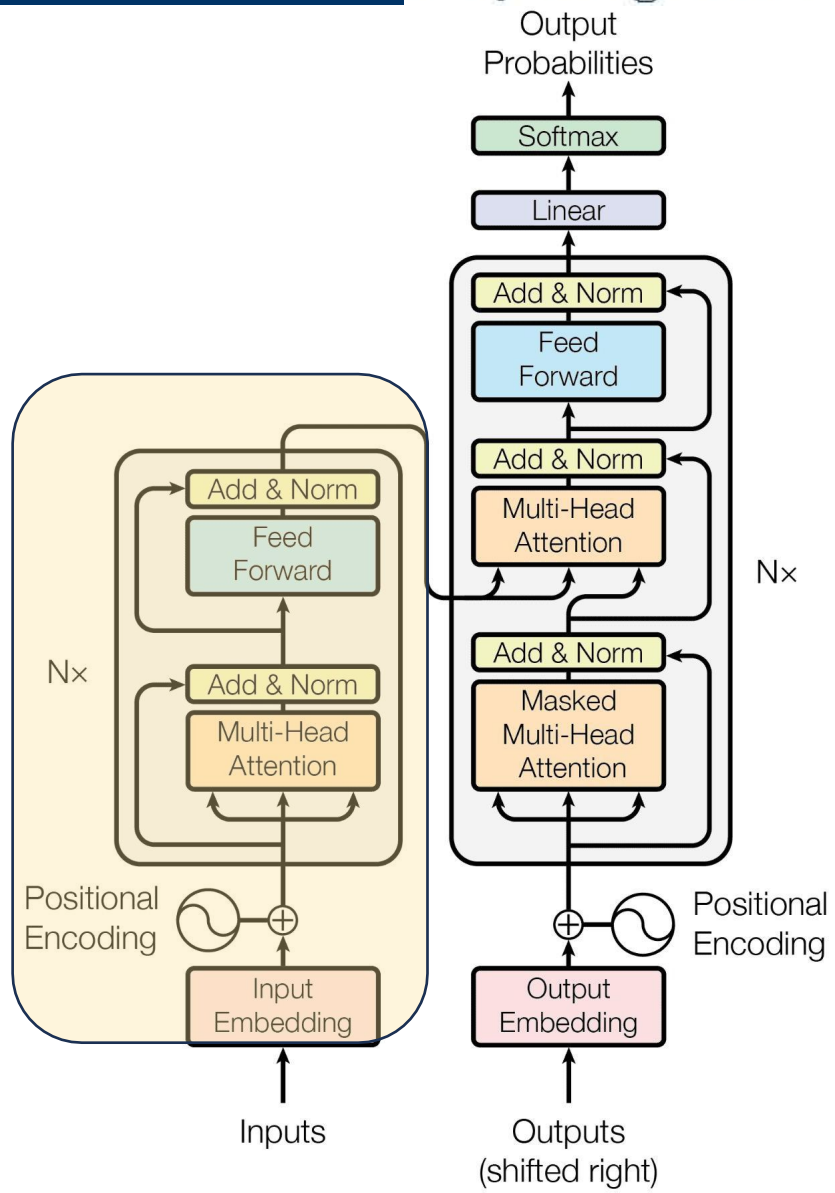


# BERT - Bidirectional Encoder Representations

## BERT Fine-Tuning:

- Simply add a task-specific module after the last encoder layer to map it to the desired dimension.
  - *Classification Tasks:*
    - *Add a feed-forward layer on top of the encoder output for the [CLS] token*
  - *Question Answering Tasks:*
    - *Train two extra vectors to mark the beginning and end of answer from paragraph*
  - *...*



# BERT - Bidirectional Encoder Representations

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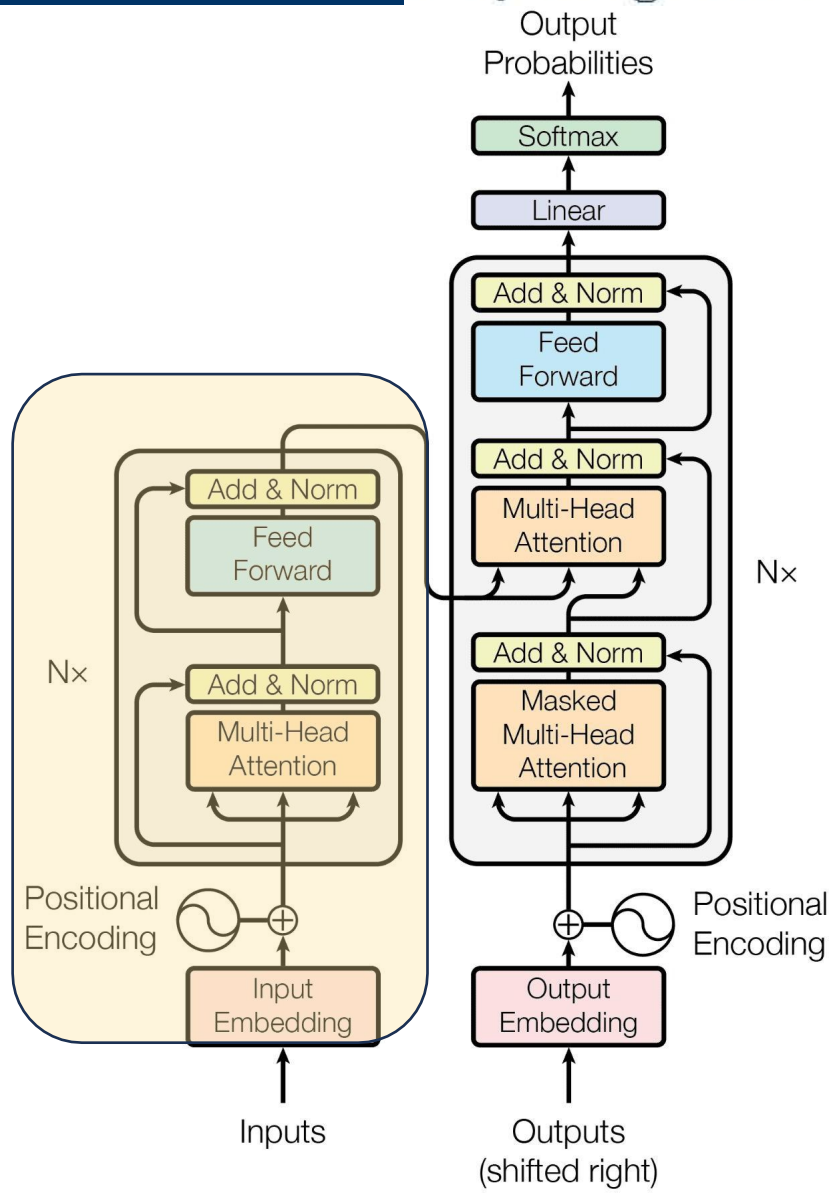
**BERT Evaluation:**

- General Language Understanding Evaluation (GLUE)
  - Sentence pair tasks
  - Single sentence classification

- Stanford Question Answering Dataset (SQuAD)



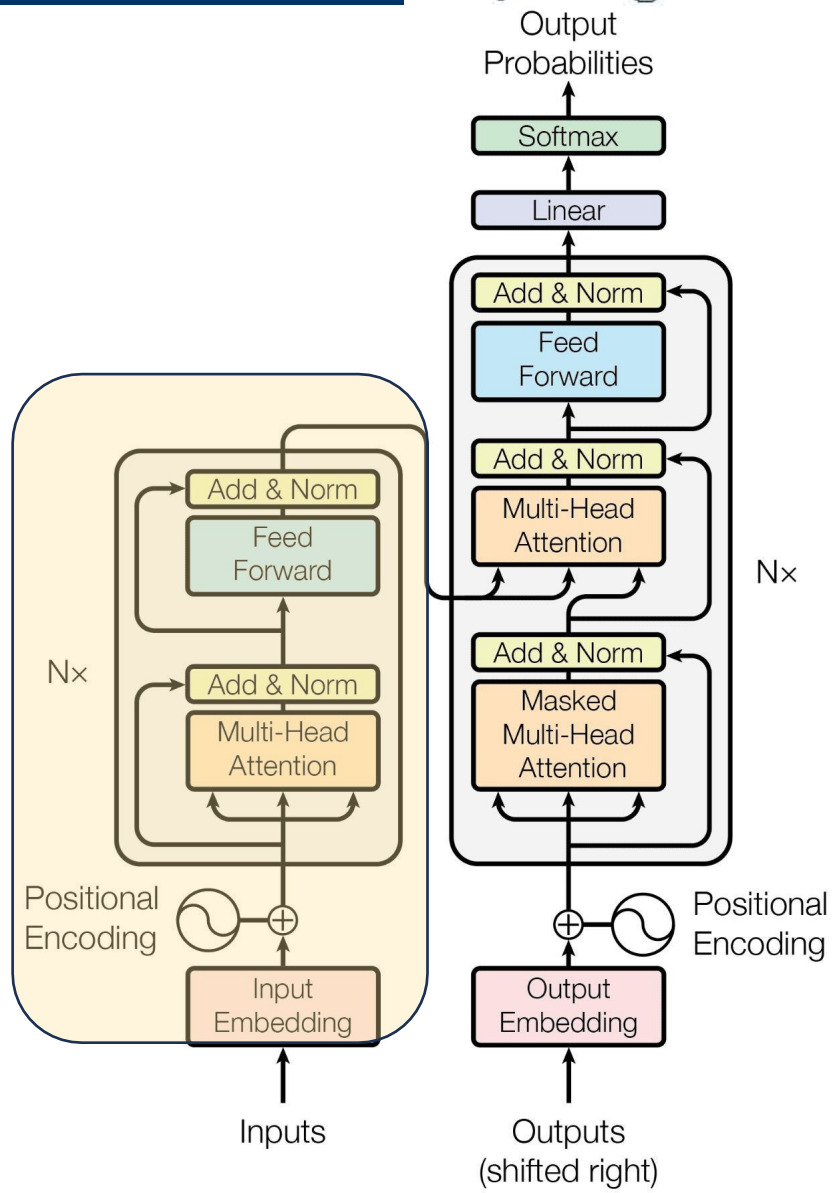
# BERT - Bidirectional Encoder Representations

## BERT Evaluation:

| System | MNLI-(m/mm) 392k | QQP 363k | QNLI 108k | SST-2 67k | CoLA 8.5k | STS-B 5.7k | MRPC 3.5k | RTE 2.5k | Average - |
|---|---|---|---|---|---|---|---|---|---|
| Pre-OpenAI SOTA | 80.6/80.1 | 66.1 | 82.3 | 93.2 | 35.0 | 81.0 | 86.0 | 61.7 | 74.0 |
| BiLSTM+ELMo+Attn | 76.4/76.1 | 64.8 | 79.8 | 90.4 | 36.0 | 73.3 | 84.9 | 56.8 | 71.0 |
| OpenAI GPT | 82.1/81.4 | 70.3 | 87.4 | 91.3 | 45.4 | 80.0 | 82.3 | 56.0 | 75.1 |
| $BERT_{BASE}$ | 84.6/83.4 | 71.2 | 90.5 | 93.5 | 52.1 | 85.8 | 88.9 | 66.4 | 79.6 |
| $BERT_{LARGE}$ | **86.7/85.9** | **72.1** | **92.7** | **94.9** | **60.5** | **86.5** | **89.3** | **70.1** | **82.1** |

| System | Dev EM | Dev F1 | Test EM | Test F1 |
|---|---|---|---|---|
| Leaderboard (Oct 8th, 2018) | | | | |
| Human | - | - | 82.3 | 91.2 |
| #1 Ensemble - nlnet | - | - | 86.0 | 91.7 |
| #2 Ensemble - QANet | - | - | 84.5 | 90.5 |
| #1 Single - nlnet | - | - | 83.5 | 90.1 |
| #2 Single - QANet | - | - | 82.5 | 89.3 |
| Published | | | | |
| BiDAF+ELMo (Single) | - | 85.8 | - | - |
| R.M. Reader (Single) | 78.9 | 86.3 | 79.5 | 86.6 |
| R.M. Reader (Ensemble) | 81.2 | 87.9 | 82.3 | 88.5 |
| Ours | | | | |
| $BERT_{BASE}$ (Single) | 80.8 | 88.5 | - | - |
| $BERT_{LARGE}$ (Single) | 84.1 | 90.9 | - | - |
| $BERT_{LARGE}$ (Ensemble) | 85.8 | 91.8 | - | - |
| $BERT_{LARGE}$ (Sgl.+TriviaQA) | **84.2** | **91.1** | **85.1** | **91.8** |
| $BERT_{LARGE}$ (Ens.+TriviaQA) | **86.2** | **92.2** | **87.4** | **93.2** |

Table 2: SQuAD results. The BERT ensemble is 7x systems which use different pre-training checkpoints and fine-tuning seeds.



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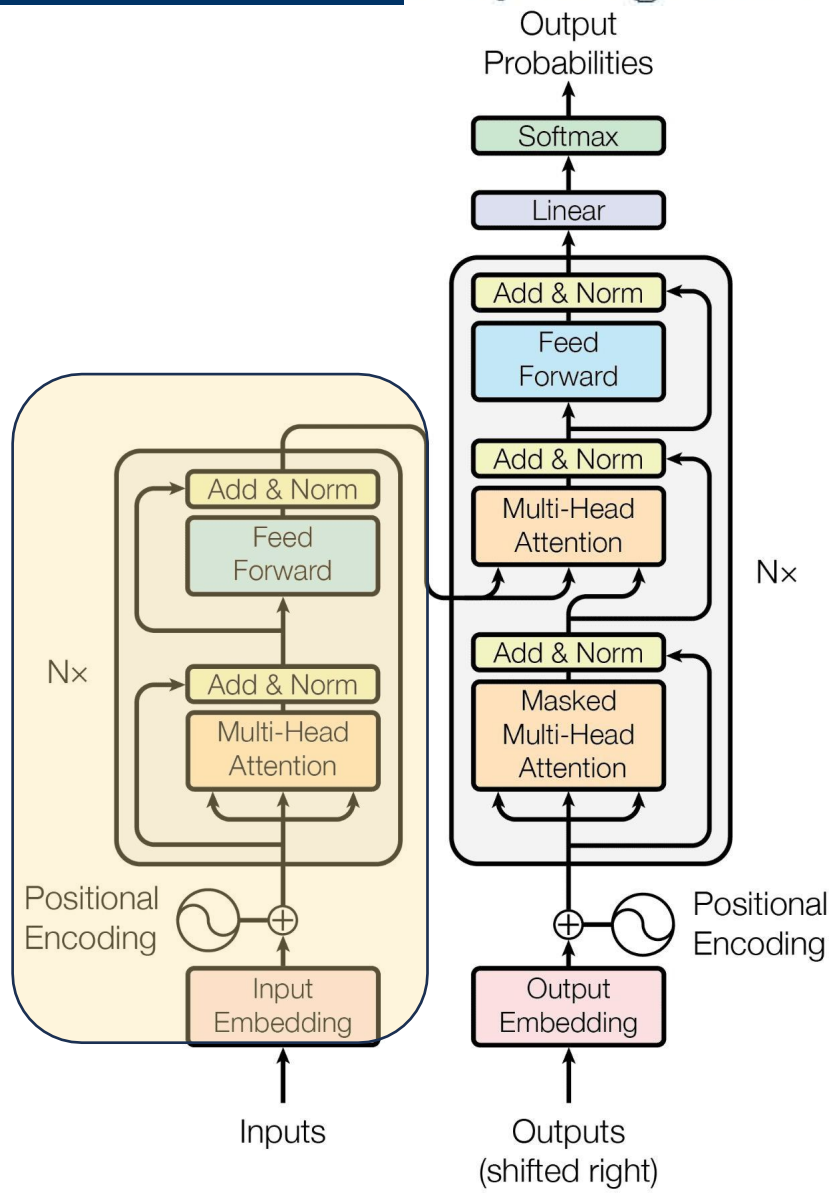
- **Pre-training tasks can be invented flexibly…**
  - Effective representations can be derived from a flexible regime of pre-training tasks.



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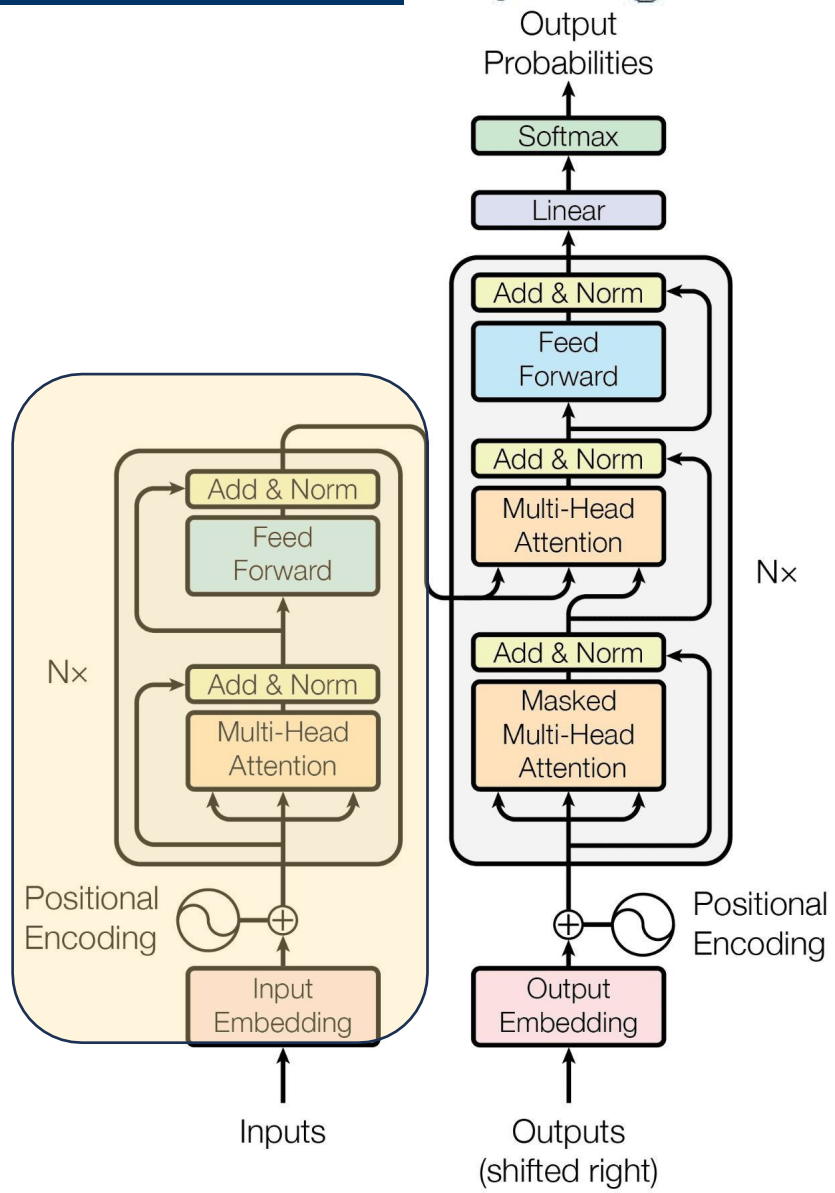
  - As long as we have effective representations, that seems to form a general model which can serve as the backbone for many specialized models.



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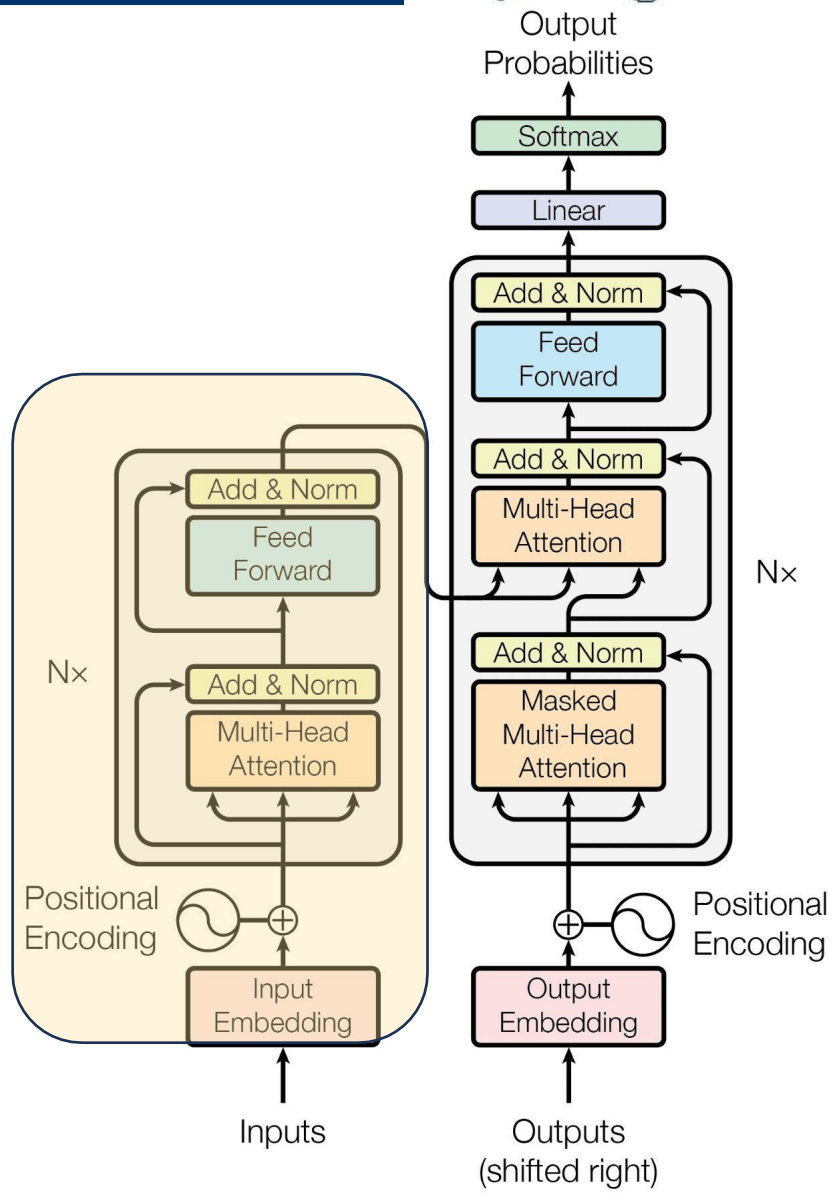
  - As long as we have effective representations, that seems to form a general model which can serve as the backbone for many specialized models.
- **And scaling works!!!**
  - 340M was considered large in 2018

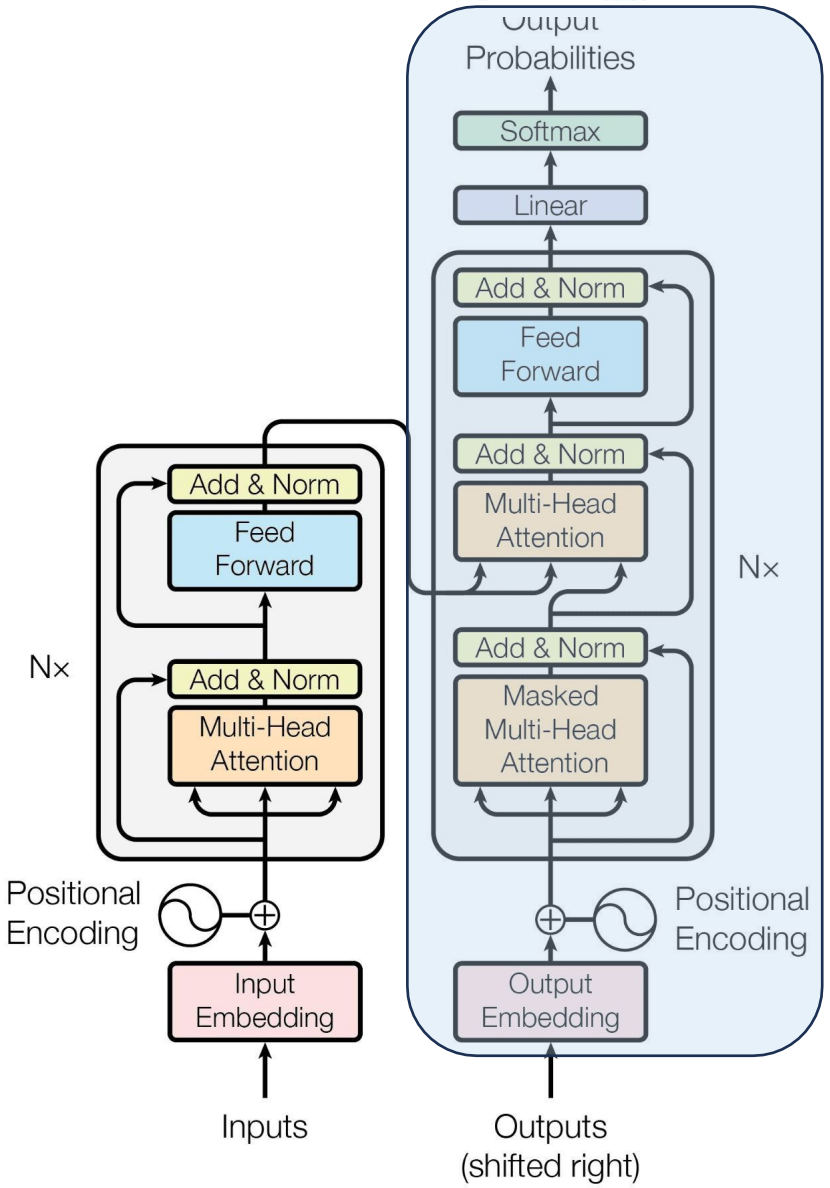


# BERT and GPT Concepts

- **GPT (Generative Pre-trained Transformer)**
  - Uses transformer decoder only.
  - Trained with next-token prediction.
  - **Unidirectional:** Considers left context only.
  - Fine-tuned for text generation, dialogue systems.
  - **Architecture:**
    - Layers of decoder blocks.
    - Causal masking to prevent future token access.
    - Self-attention heads capture sequential dependencies.

# GPT – Generative Pretrained Transformer

- Similarly motivated as BERT, though differently designed
  - Can we leverage large amounts of unlabeled data to pretrain an LM that understands general patterns?



# GPT – Generative Pretrained Transformer

**GPT Pre-Training Corpus:**

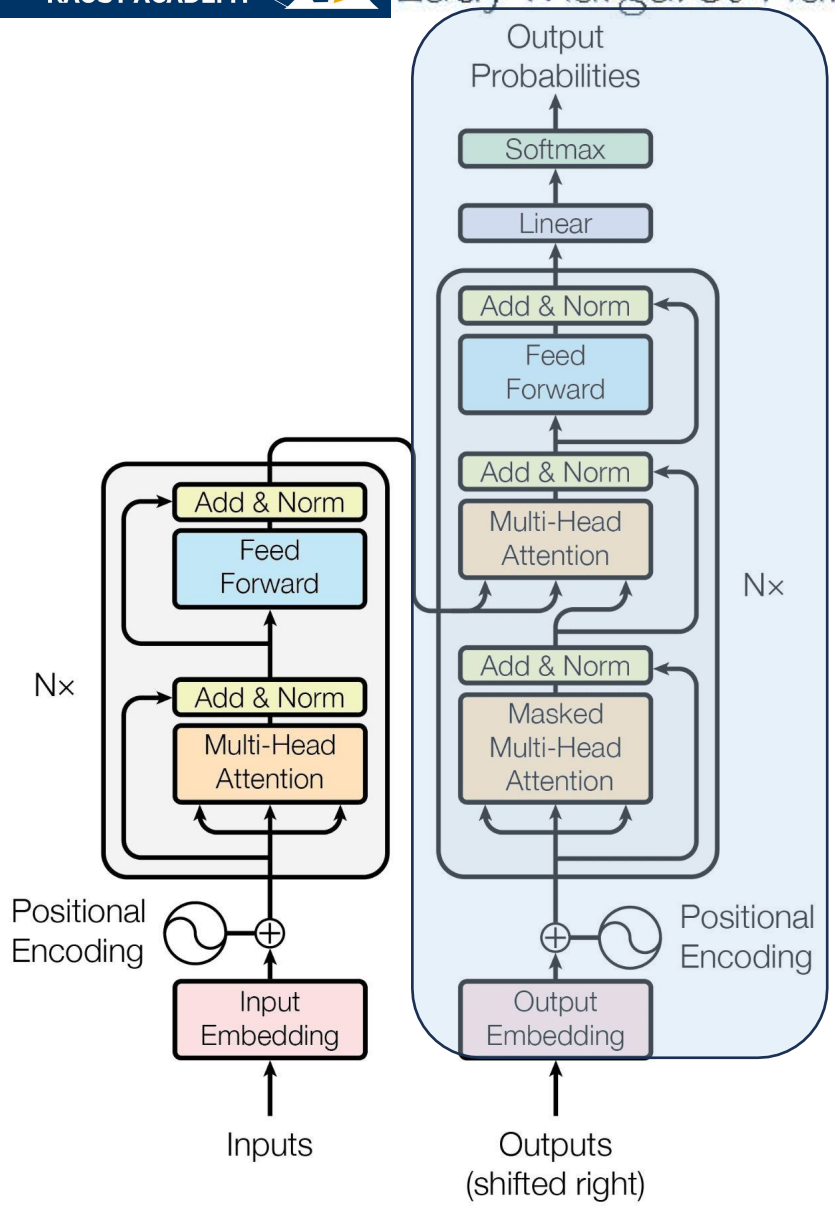
- Similarly, BooksCorpus and English Wikipedia

**GPT Pre-Training Tasks:**

- Predict the next token, given the previous tokens
  - More learning signals than MLM

**GPT Pre-Training Results:**

- GPT – 117M Params
  - Similarly competitive on GLUE and SQuAD



# GPT – Generative Pretrained Transformer

## GPT Fine-Tuning:

- Prompt-format task-specific text as a continuous stream for the model to fit

**Summarization**

Summarize this article:

The summary is:

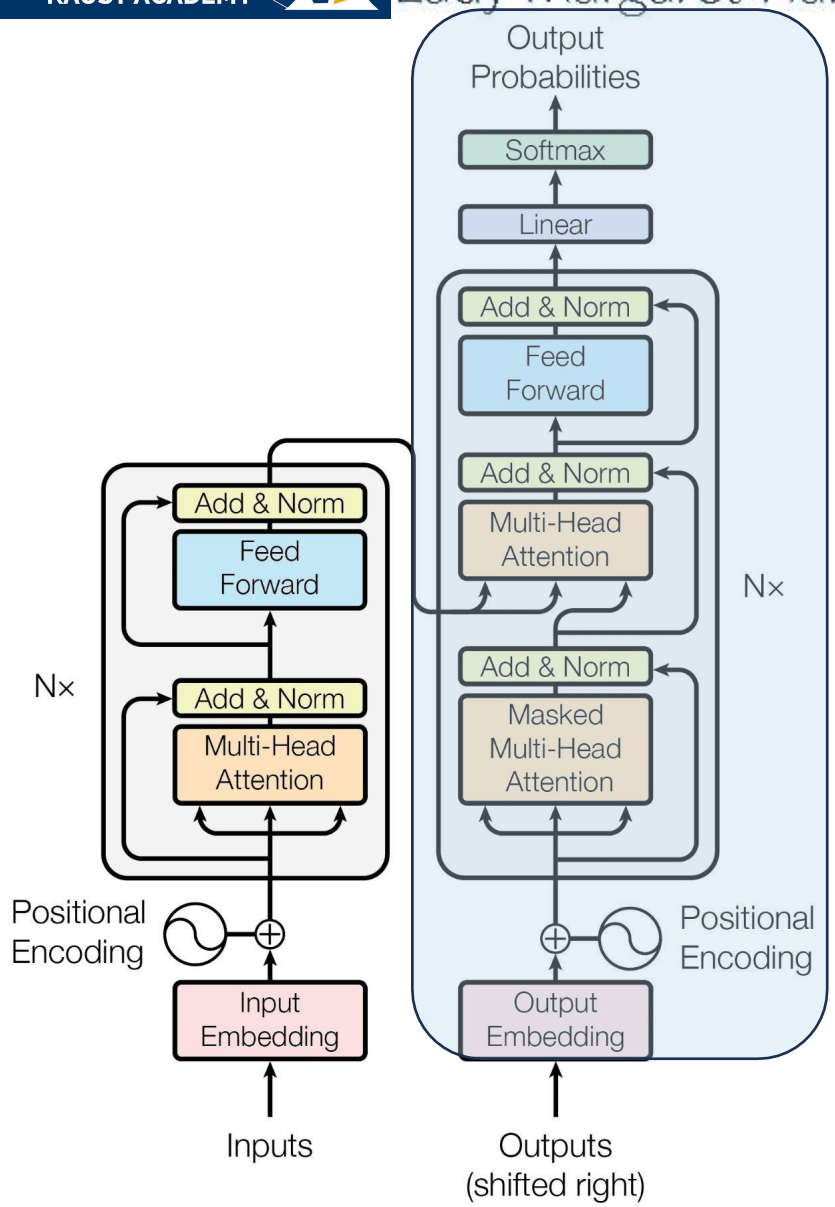
**QA**

Answer the question based on the context.

Context:

Question:

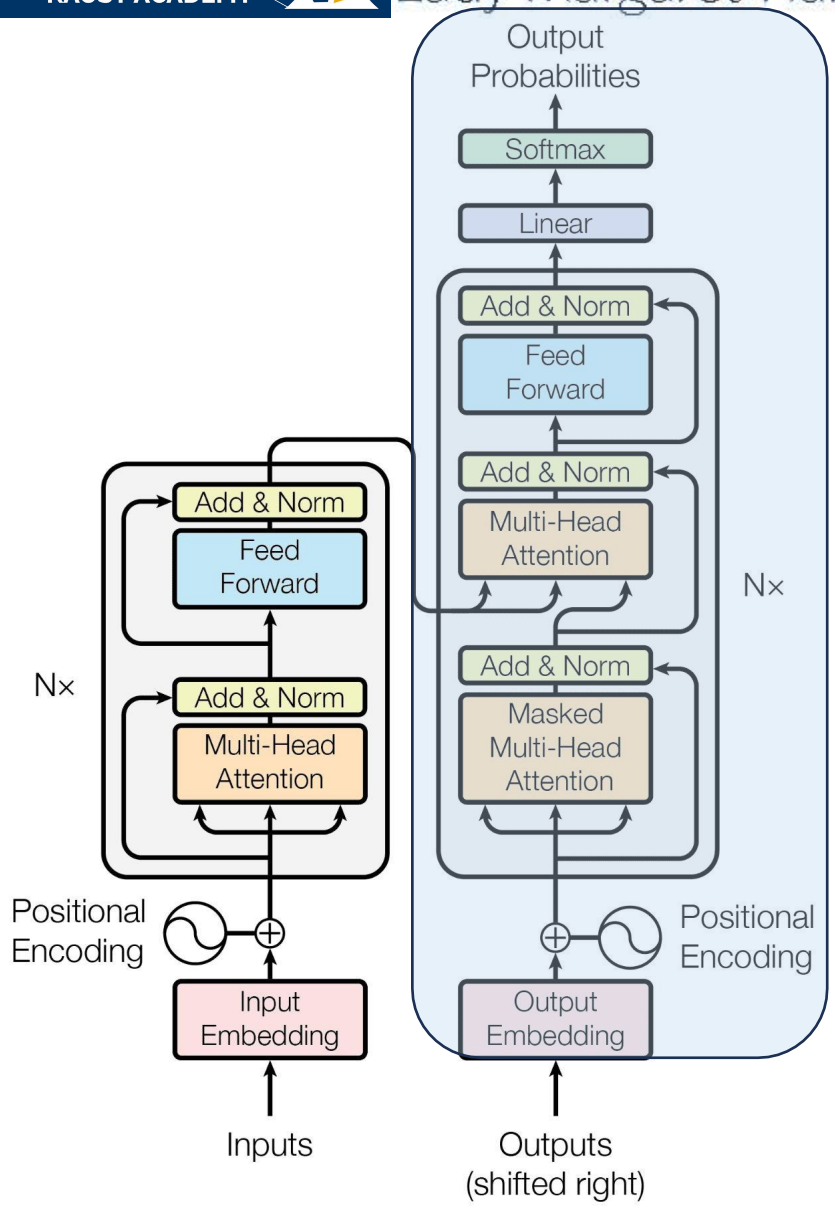
Answer:



# GPT – Generative Pretrained Transformer

## What is our takeaway from GPT?

- **The Effectiveness of Self-Supervised Learning**
  - Specifically, the model seems to be able to learn from generating the language *itself*, rather than from any specific task we might cook up.



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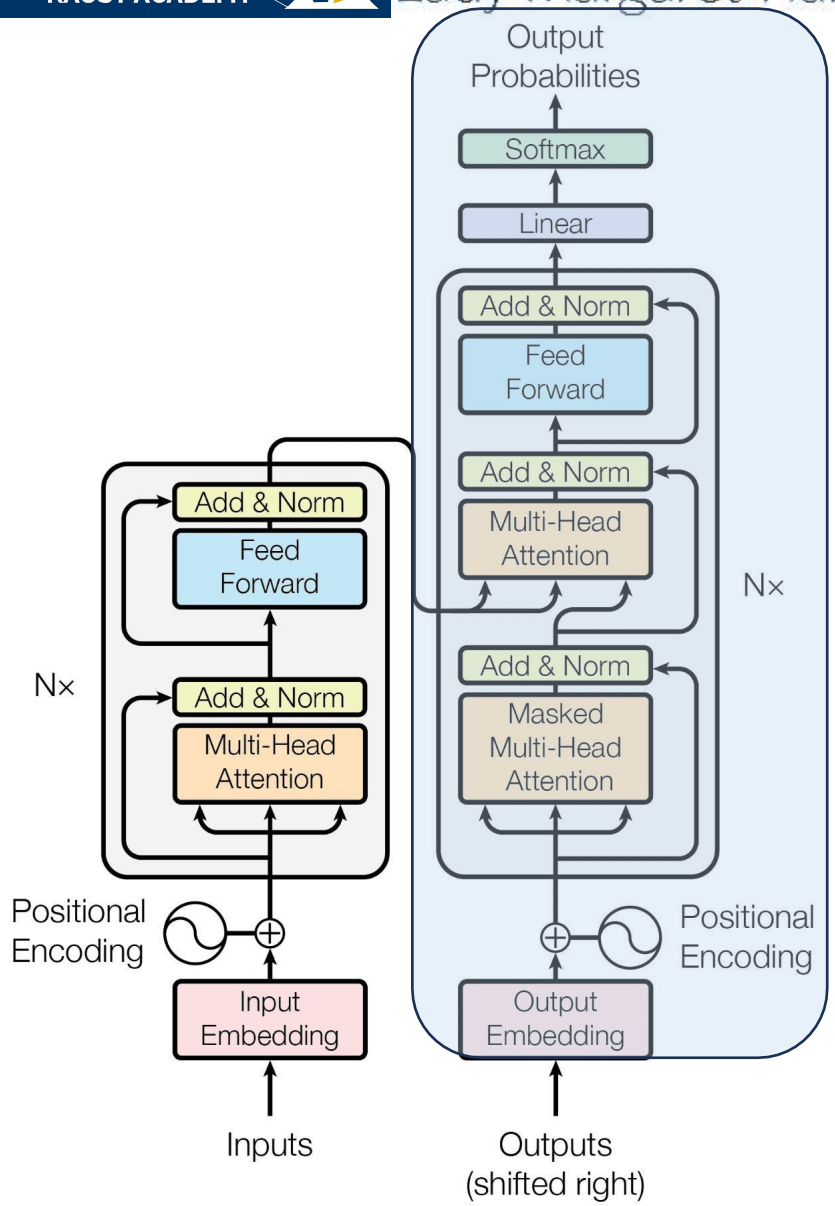
  - Specifically, a generatively pretrained model seems to have a decent zero-shot performance on a range of NLP tasks.



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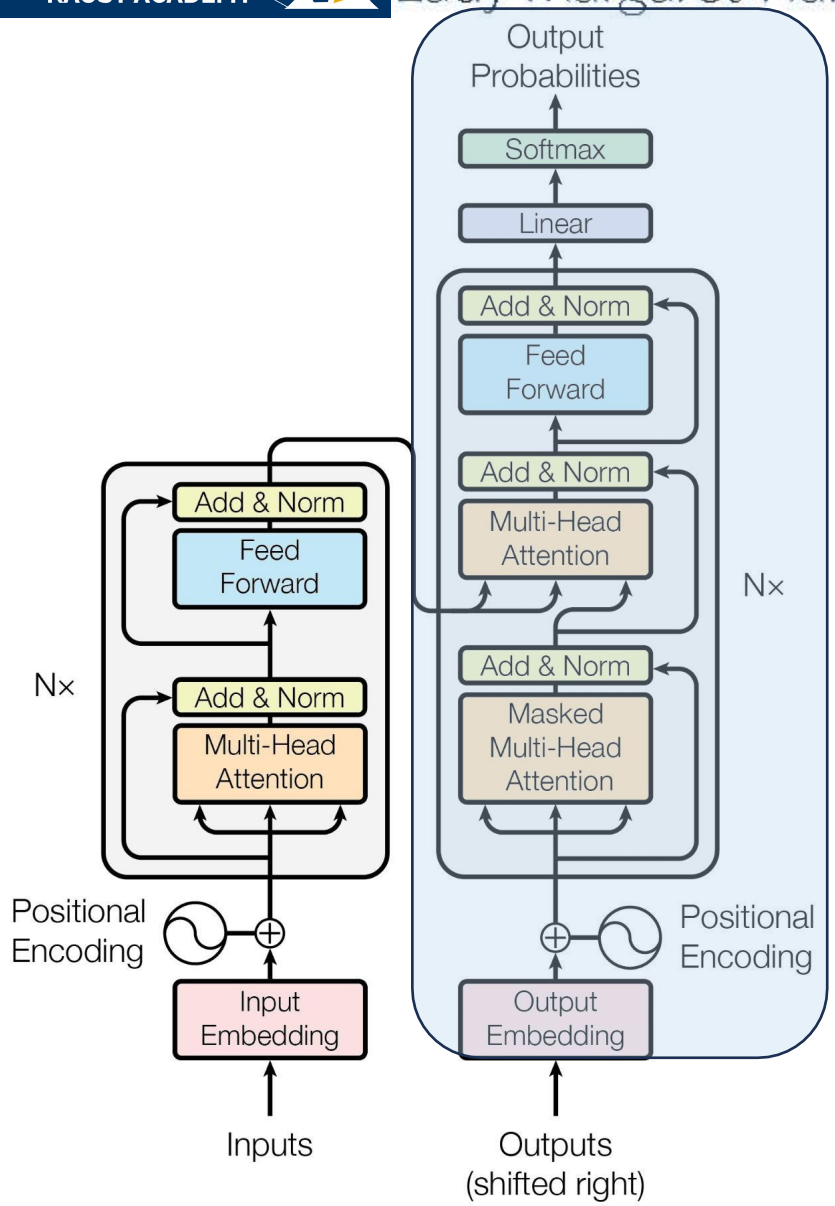
- **And scaling works!!!**



# BERT and GPT Concepts

- **Key Differences:**

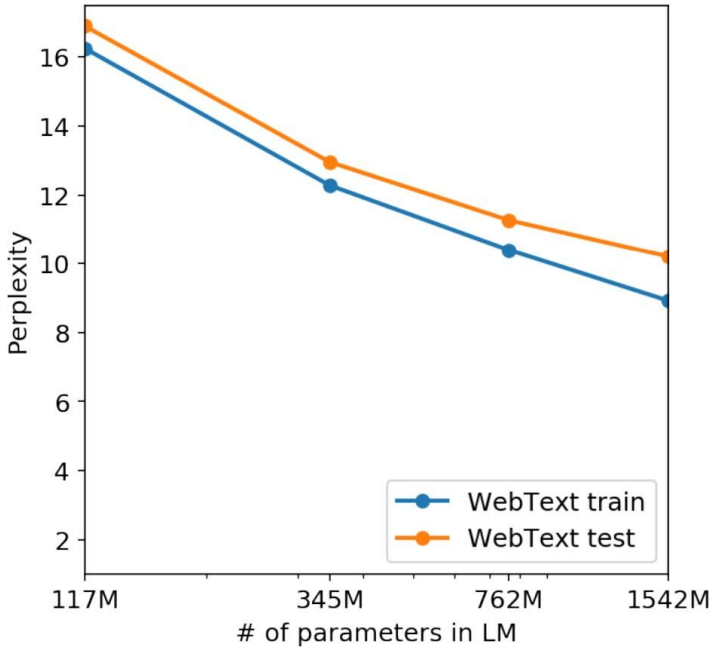
| Feature | BERT | GPT |
|---|---|---|
| Directionality | Bidirectional | Unidirectional |
| Objective | Masked Language Modeling (MLM) | Causal Language Modeling (CLM) |
| Output | Contextual embeddings | Text generation |
| Usage | Downstream tasks (e.g., classification, QA) | Generation, few-shot learning |

# Scaling Laws for LLMs

- **Kaplan et al. (2020):** "Scaling Laws for Neural Language Models"
- Performance improves predictably with:
  - More parameters
  - More compute
  - Larger datasets
- **Optimal allocation of compute:** Train bigger models with less data, rather than small models with lots of data.
- **Implication:** LLMs like GPT-3 (175B), GPT-4 (est. >500B) are products of scaling laws.

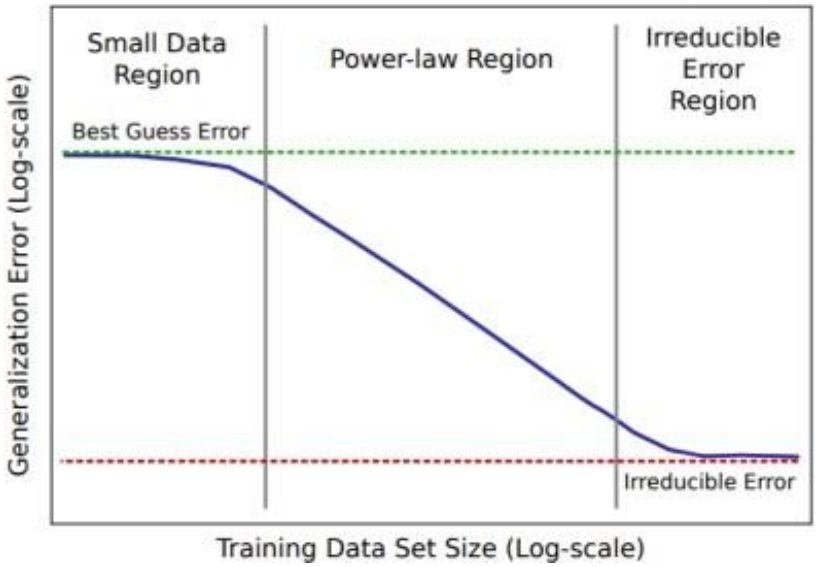
# Scaling in GPT-2

- Scaling improves the perplexity of the LM and improves performance



# Why is this interesting? Look at data scaling

- We know that typical scaling effects look like this when we increase the amount of training data



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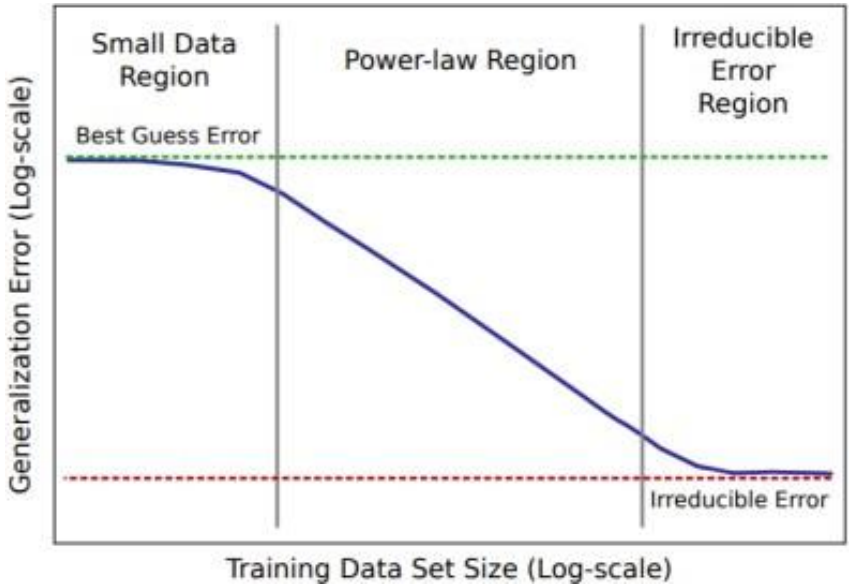
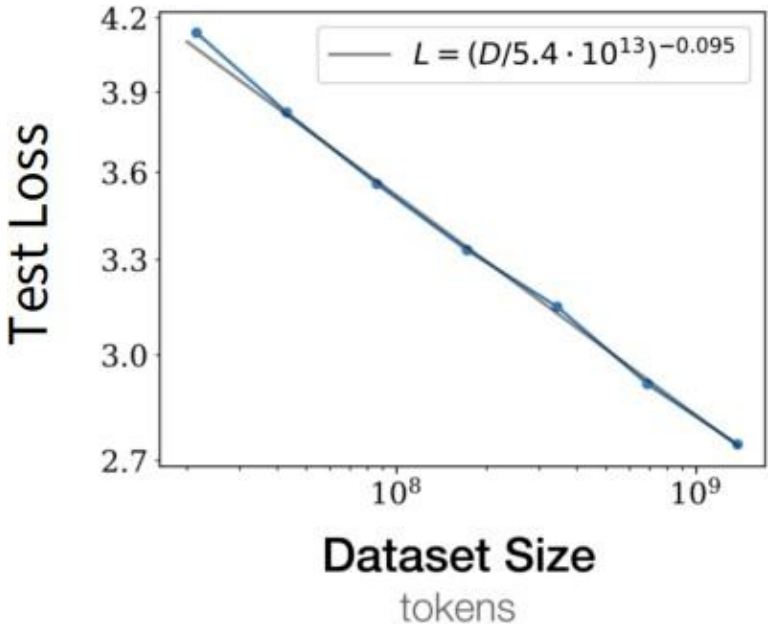
- Loss and dataset size is linear on a log-log plot
- This is "power-law scaling"



# Scaling - (Kaplan,2020)

- Can we understand scaling by positing scaling laws ?
- With scaling laws, we can make decisions on architecture, data, hyperparameters by training smaller models
- Open AI Study : **Scaling Laws for Neural Language Models** (Kaplan et al. 2020)

# Scaling - (Kaplan,2020)

- Open AI Study : **Scaling Laws for Neural Language Models** (Kaplan et al. 2020)

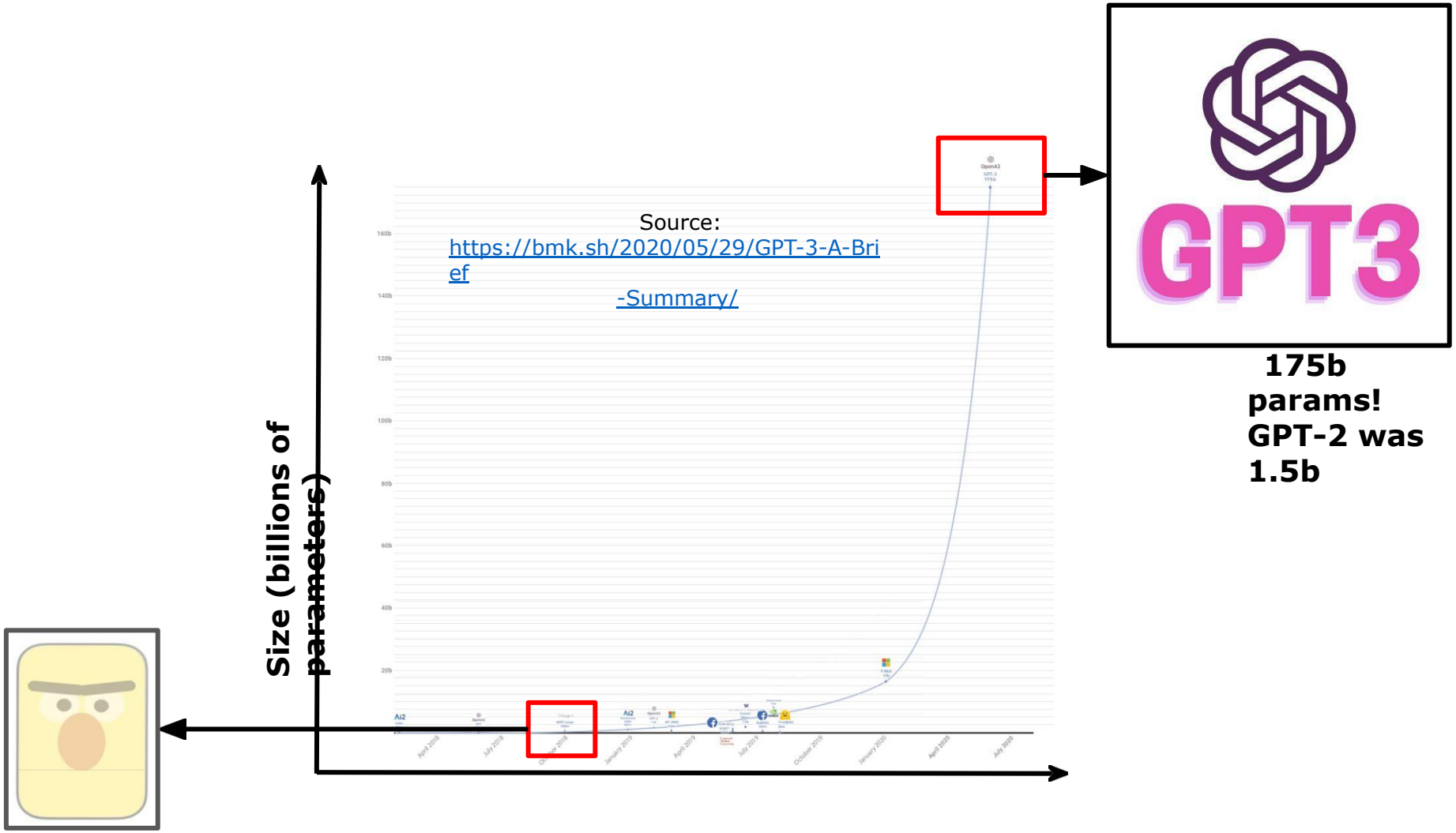
- Key Findings:
  - Performance depends strongly on scale, and weakly on the model shape
  - Larger models are more sample-efficient
  - Smooth power laws ($y = ax^k$) b/w empirical performance & N - parameters, D - dataset size, C - compute

# Scaling Effects

- The effect of some hyperparameters on big LMs can be predicted before training – optimizer (Adam v/s SGD), model depth, LSTM v/s Transformer

- Idea:
  - Train a few smaller models
  - Establish a scaling law (e.g. ADAM vs SGD scaling law)
  - Select optimal hyper param based on the scaling law prediction

# Model Scaling: GPT-3



# Emergent Abilities with GPT-3 – Wei et. al 2022

- Emergent abilities:
  - not present in smaller models but is present in larger models
  - Do LLMs like GPT3 have these ?

- Findings:
  - GPT-3 trained on text can do arithmetic problems like addition and subtraction
  - Different abilities “emerge” at different scales

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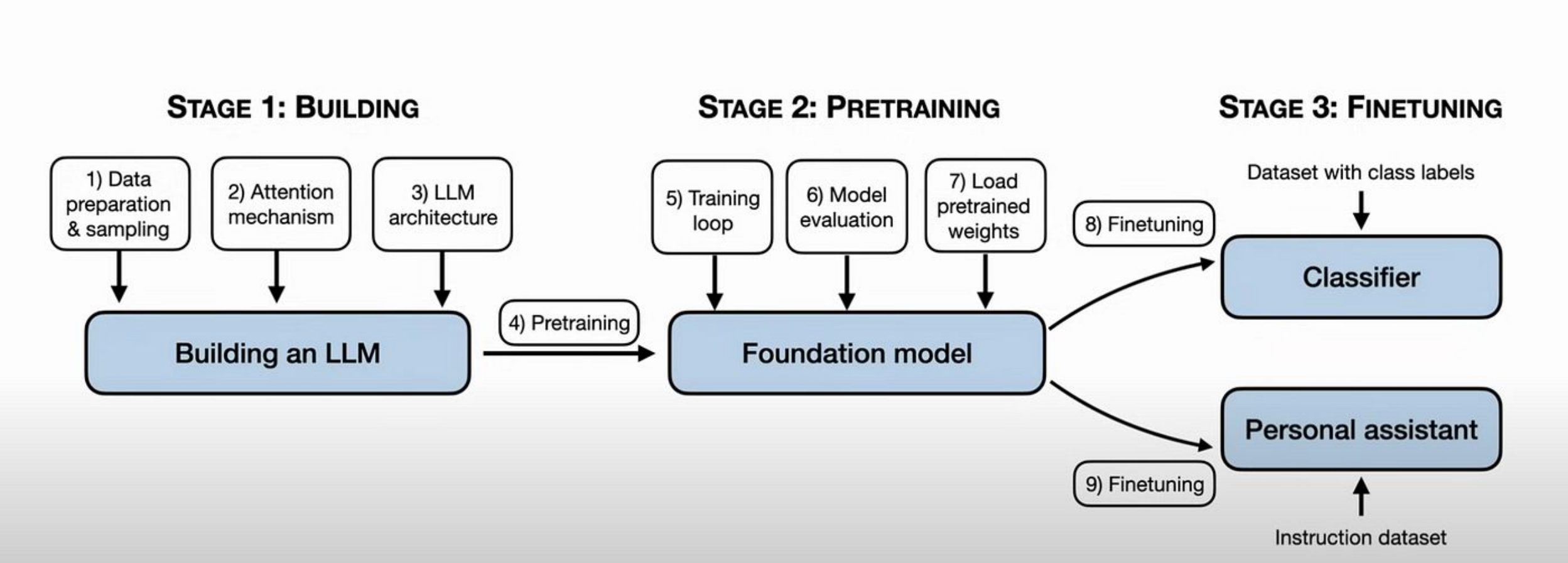
- Findings:
  - GPT-3 trained on text can do arithmetic problems like addition and subtraction
  - Different abilities “emerge” at different scales
  - Model scale is not the only contributor to emergence – for 14 BIG-Bench tasks, LaMDA 137B and GPT-3 175B models perform at near-random, but PaLM 62B achieves above-random performance
  - Problems LLMs can’t solve today may be emergent for future LLMs

# Pre-training Overview

- **Pre-training Phase:**
  - Large-scale unsupervised training on corpus (e.g., Common Crawl, Books)
  - Objective: learn general-purpose language representations
- **Why Pre-train?**
  - Data-efficient fine-tuning
  - Enables zero-shot and few-shot capabilities
  - Foundation for instruction tuning, alignment
- **Challenges:**
  - Massive compute costs
  - Environmental concerns (carbon footprint)

# Pre-training Overview



# Training of Decoder-only LLMs – Llama 2

1. Auto-regressive Pre-training - Train to predict the next token on very large-scale corpora ( ~3 trillion tokens)

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1. Auto-regressive Pre-training - Train to predict the next token on very large scale corpora ( ~3 trillion tokens)
2. Instruction Fine-tuning/ Supervised Fine-tuning (SFT) - Fine-tune the pre-trained model with pairs of (instruction+input,output) with large dataset and then with small high-quality dataset

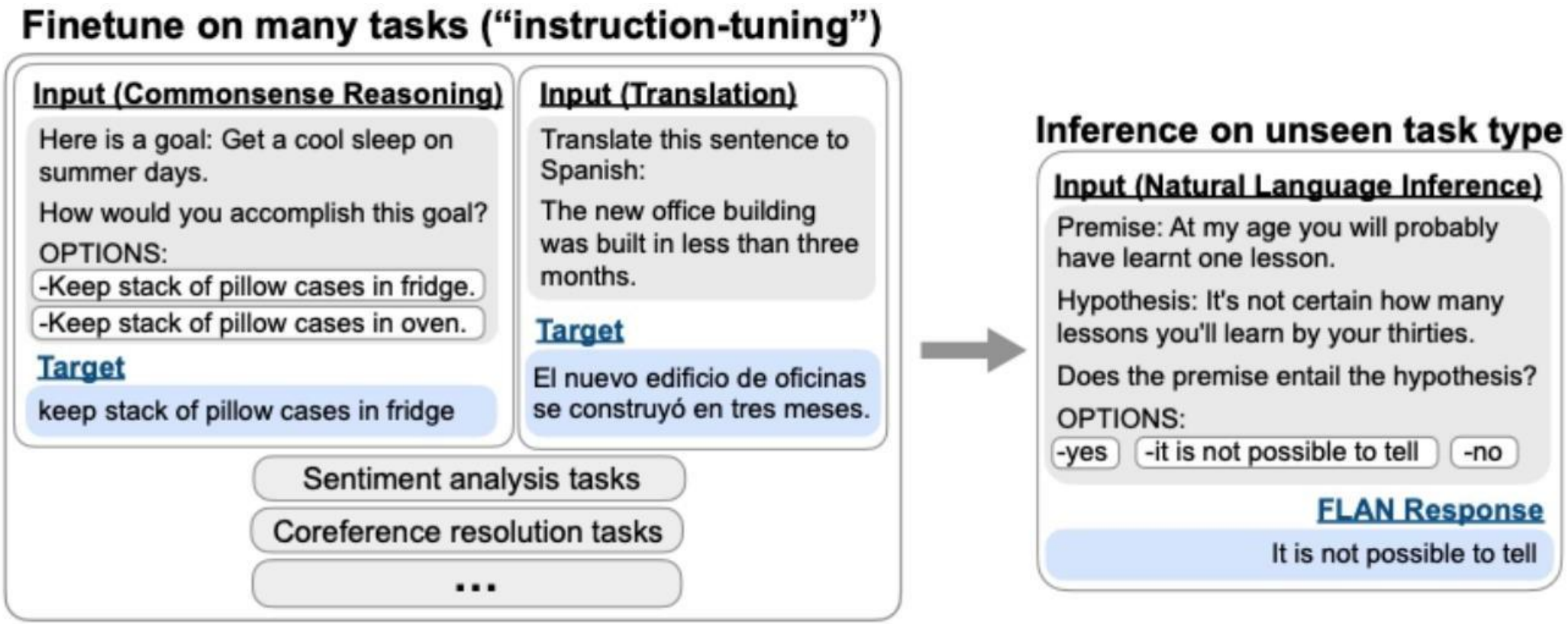
Instruction fine-tuning provides as a prefix a natural language description of the task along with the input.

- E.g. Translate into French this sentence: my name is -> je m'appelle

# Supervised Fine-tuning versus Pre-training

- Objective function
  - Loss computed only for target tokens in SFT, all tokens are targets in pre-training
- Input and Target
  - Instruction + input as input with the target in SFT and only input as input with shifted input as target
- Purpose
  - Pre-training makes good generalist auto-completes but good SFT builds models that can do many unseen tasks
  - SFT can also guide nature of outputs in terms of safety and helpfulness

# Instruction Tuning (Wei et. al. 2021)



# Fine-Tuning Methods for LLMs

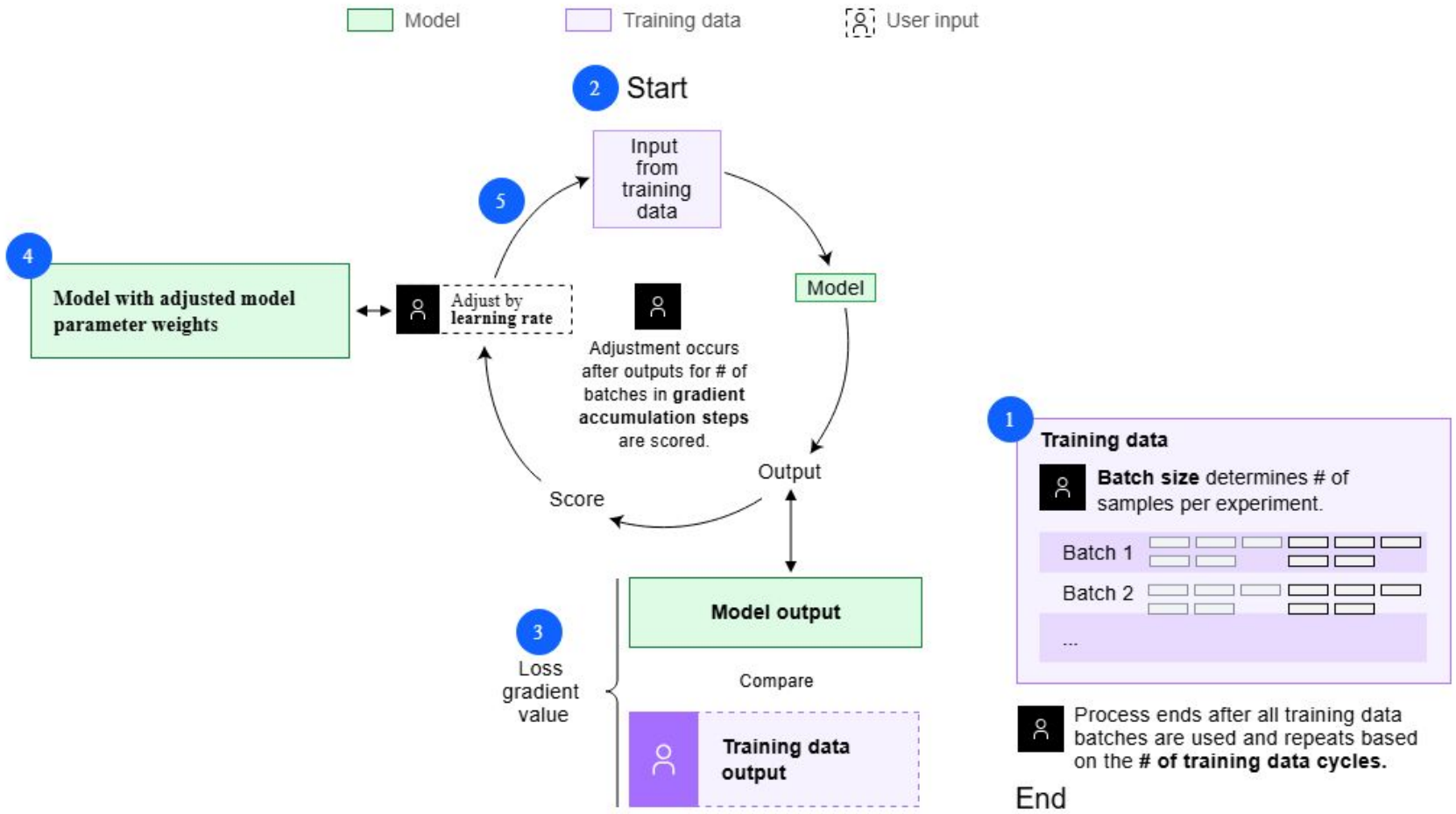
# Categories of Fine-Tuning

| Method | Description | Parameters Updated | Efficiency |
|---|---|---|---|
| Full Fine-Tuning | Retrain all model weights | All | Expensive |
| Adapter Tuning | Add small bottlenecks (e.g., Houlsby) | Few | Efficient |
| Prefix Tuning | Tune soft prompts | Few tokens | Efficient |
| LoRA / QLoRA | Low-rank decomposition of weight deltas | Very few | Very Efficient |
| Instruction Tuning | Fine-tune on instruction-following datasets | All or partial | |

# Full Fine-Tuning (FT)

- **Definition:** Fine-tune all parameters of the pre-trained model on downstream data.
- **Historical Use:** Common in early GPT-2 and BERT applications.
- **Pros:**
  - Maximum flexibility and performance
- **Cons:**
  - Expensive (requires large compute resources)
  - Prone to catastrophic forgetting
  - Not efficient for large models

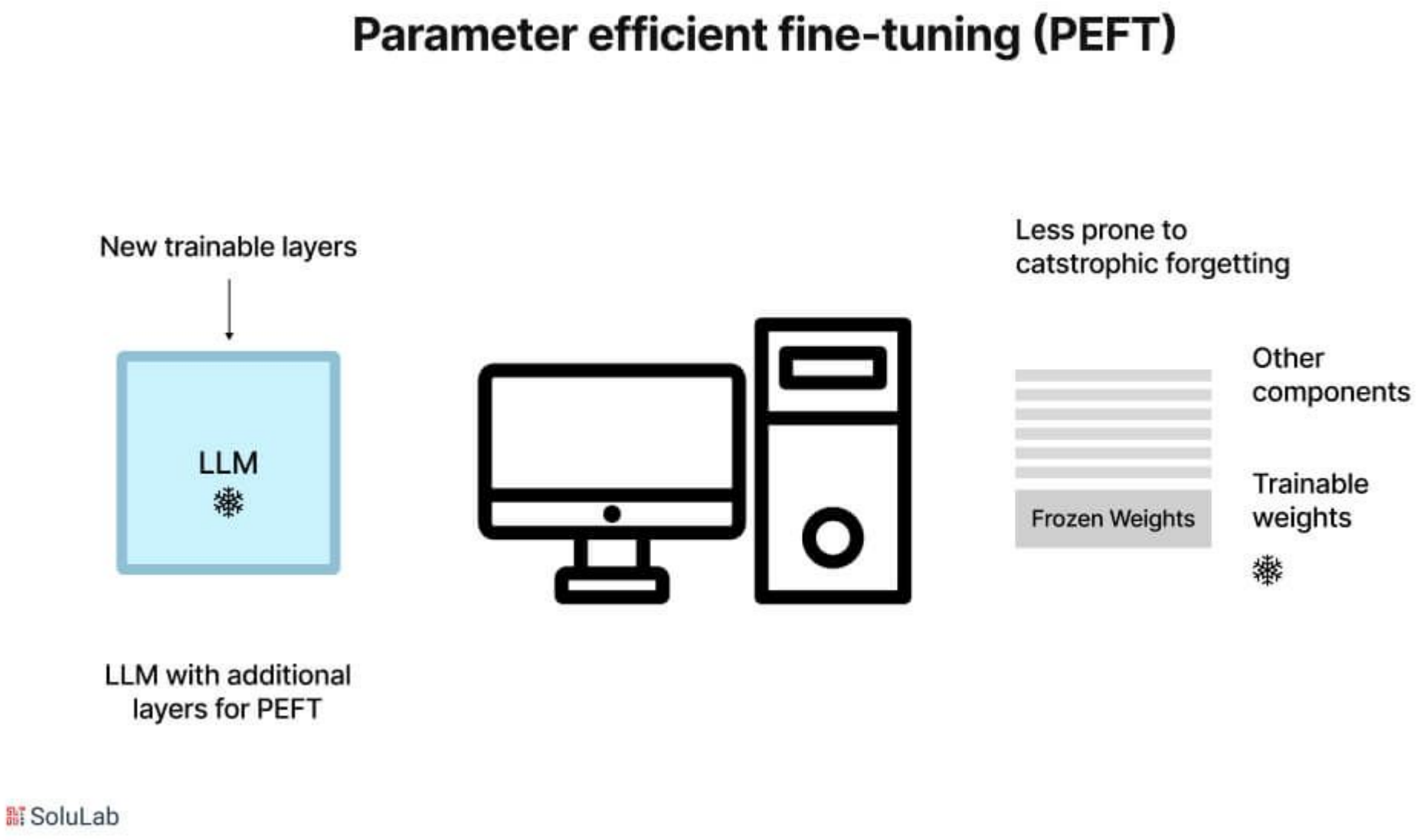
# Full fine-tuning workflow



# Parameter-Efficient Fine-Tuning (PEFT)

- **Key Idea:** Keep the base model frozen, tune only a small subset of parameters.
- **Types:**
  - Adapter Modules (Houlsby et al., 2019)
  - Prompt Tuning / Prefix Tuning
  - LoRA / QLoRA
- **Benefits:**
  - Enables multi-tasking and personalization
  - Suitable for low-resource adaptation
  - Reduces compute and memory requirements

# Parameter-Efficient Fine-Tuning (PEFT)



# Supervised Fine-Tuning (SFT)

# What is Supervised Fine-Tuning (SFT)?

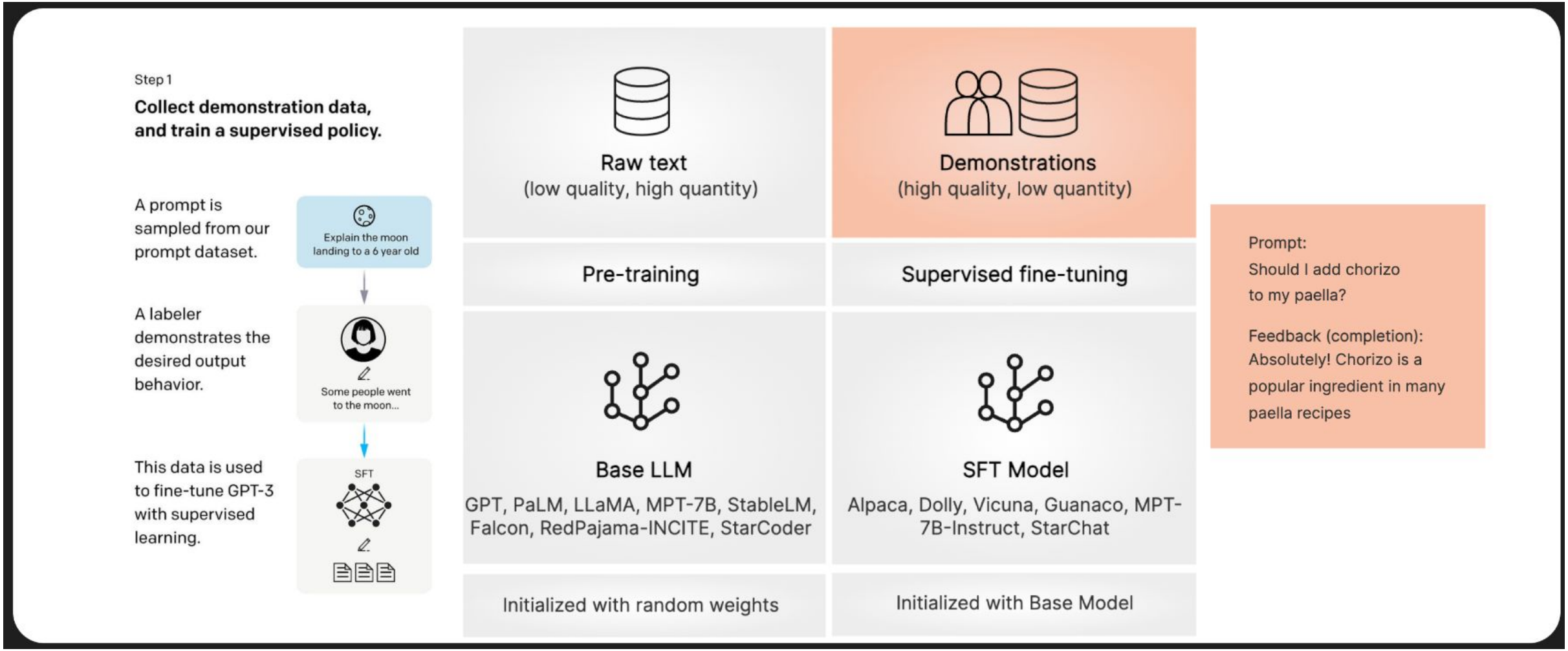
- **Definition:** SFT is training on labeled instruction-response pairs.
- **Example:**
  - *Prompt:* "Explain black holes to a 5-year-old"
  - *Response:* "Black holes are like big vacuum cleaners in space. . ."
- **Objective:** Optimize the log-likelihood of the correct response given the prompt.

**Loss Function:**

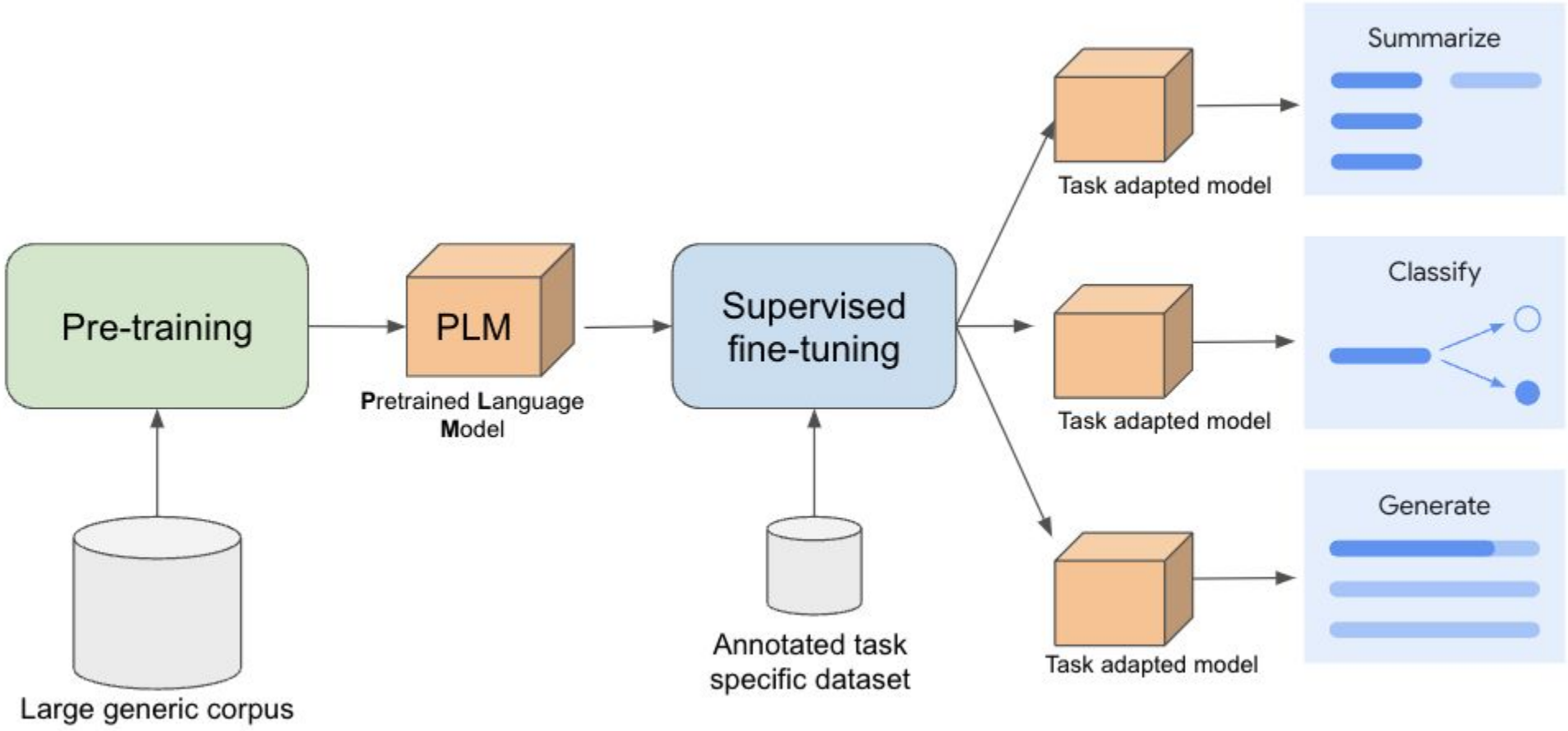
$$L_{\mathrm{SFT}} = -\sum_{t=1}^{T} \log p_\theta(y_t \mid y_{<t}, x)$$

where $x$ is the prompt, $y$ is the response, and $T$ is the response length.

# What is Supervised Fine-Tuning (SFT)?



# Supervised Fine Tuning for Gemini LLM




**Supervised Fine Tuning for Gemini LLM**

# Datasets for Supervised Fine-Tuning (SFT)

- **Instructional datasets:**
  - OpenAI: InstructGPT (Anthropic Helpfulness data)
  - Stanford Alpaca (52k GPT-3 generated instructions)
  - Dolly, ShareGPT, OASST, UltraChat
- **Note:** Quality of supervision greatly affects model behavior.

# Limitations of Supervised Fine-Tuning (SFT)

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- Cannot capture nuanced human preferences or values.
- May reinforce existing biases or hallucinations present in the data.
- Risk of overfitting, especially on synthetic or noisy datasets.
- Often leads to safe but bland and generic responses.

# LoRA and Quantized LoRA

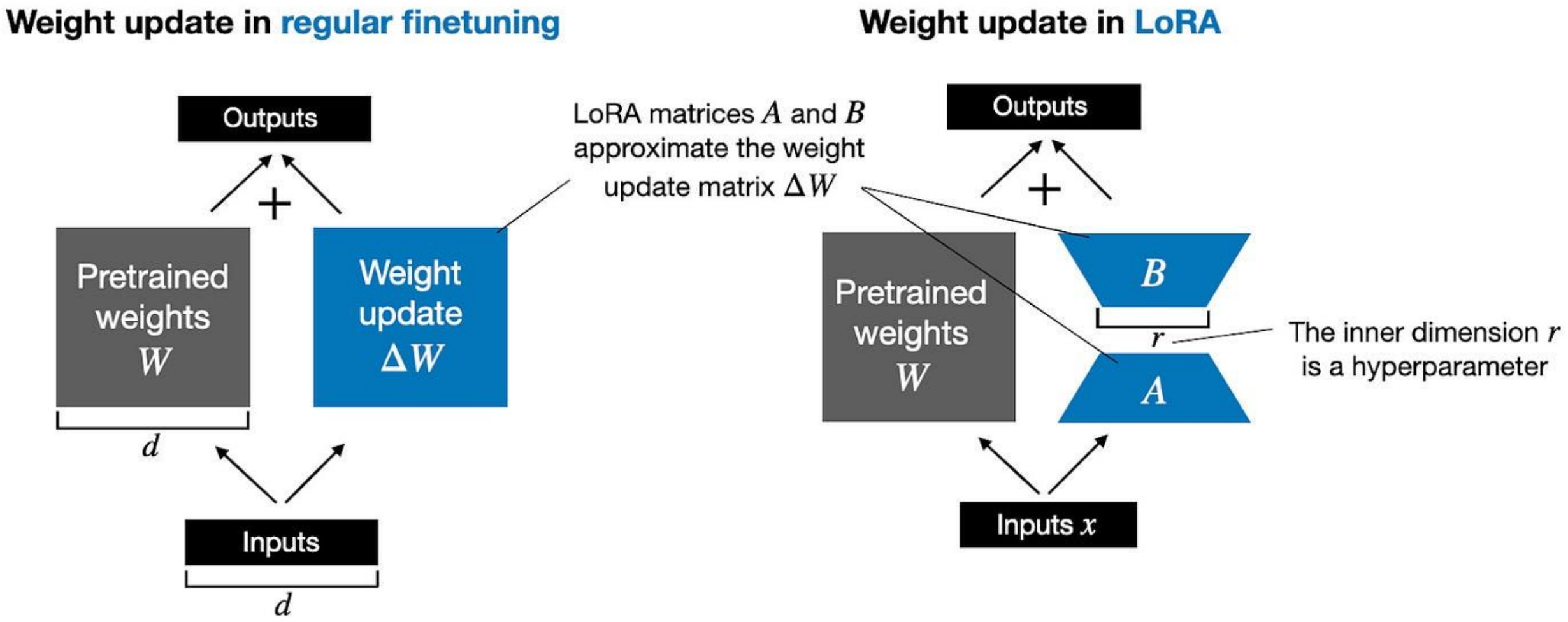
# What is LoRA?

- **Key Idea:** Update low-rank matrices instead of full weights.
- For a weight matrix $W \in \mathbb{R}^{d \times k}$:

$$W' = W + \Delta W, \quad \Delta W = AB^T$$

  where $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{k \times r}$.
- Only train $A, B \rightarrow$ drastically reduces parameters.

# What is LoRA?


Weight update in regular finetuning
Outputs
+
Pretrained weights W
Weight update ΔW
d
Inputs
d
LoRA matrices A and B approximate the weight update matrix ΔW
Weight update in LoRA
Outputs
+
Pretrained weights W
B
r
A
The inner dimension r is a hyperparameter
Inputs x

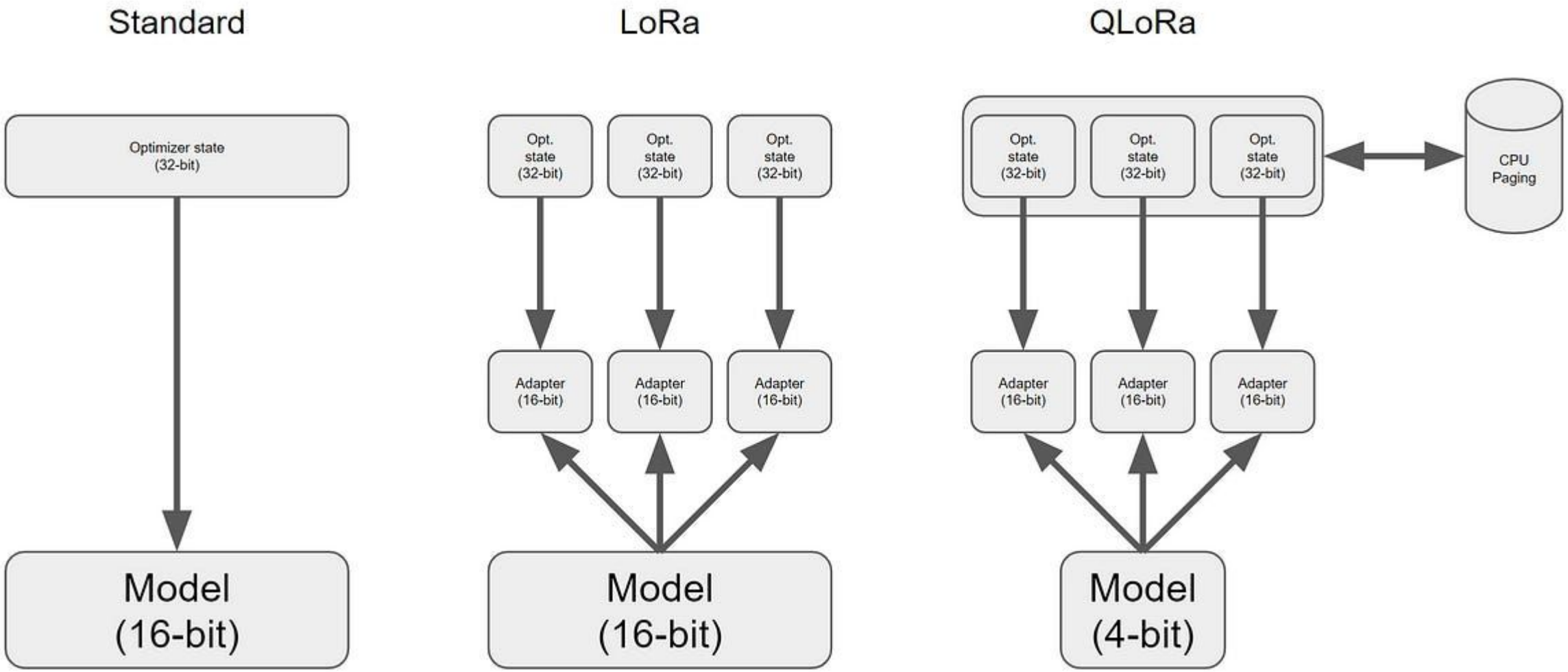
# Benefits of LoRA

- **Very lightweight:** Only 0.1%–1% of parameters are trainable.
- **Hardware friendly:** Enables fine-tuning on consumer GPUs and even laptops.
- **Modular:** Supports plug-and-play adapters for different tasks or users.
- **Personalization:** Allows efficient user- or domain-specific adaptation.

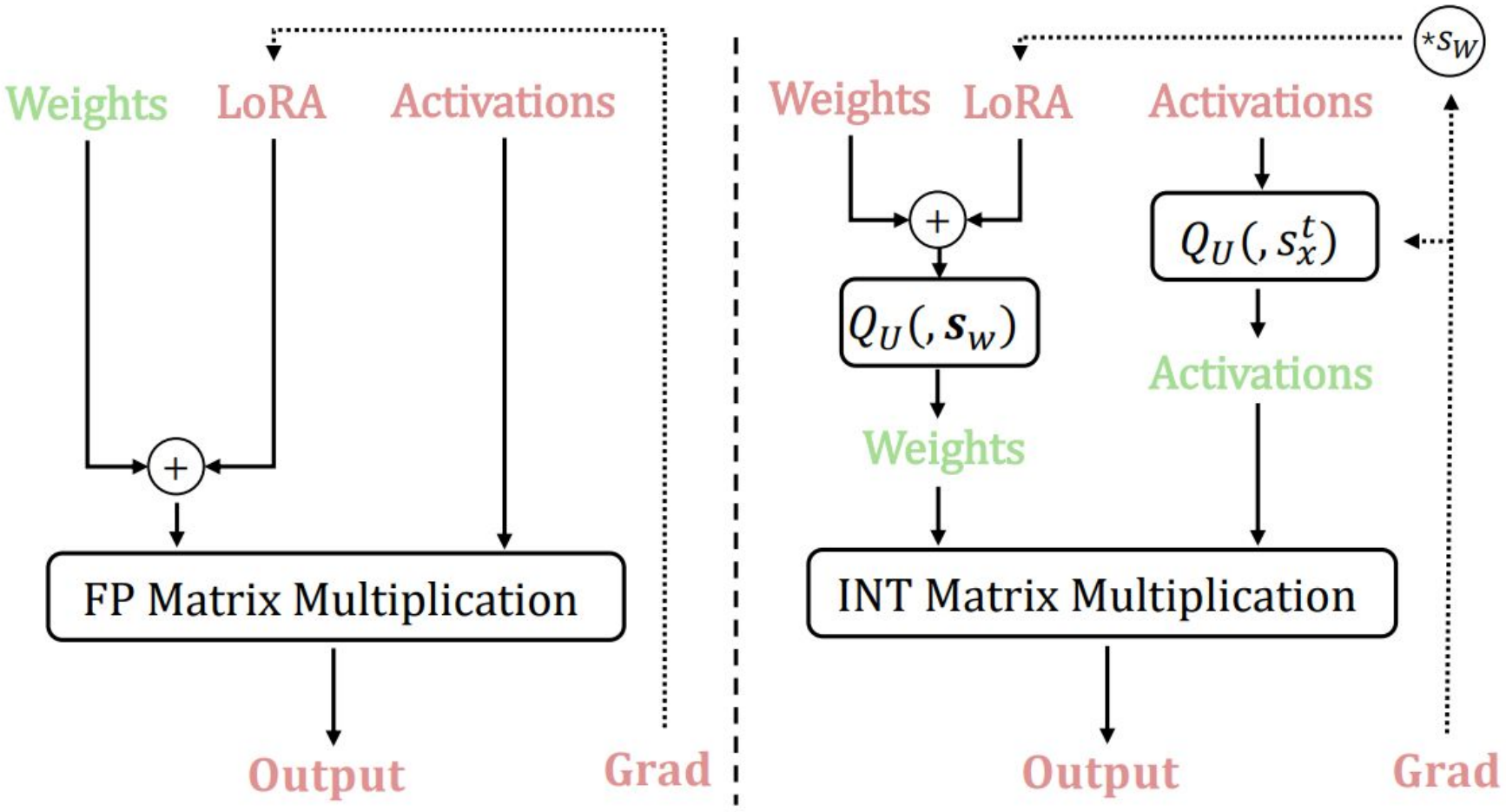
# QLoRA — Quantized LoRA

- **Key Idea:** Fine-tune large language models in 4-bit precision using LoRA adapters.
- **Scalability:** Enables fine-tuning of 65B parameter models on a single 48GB GPU.
- **Efficiency:** Highly memory-efficient with no significant performance loss (Dettmers et al., 2023).
- **Techniques:**
  - Double quantization
  - Paged optimizers
  - NF4 (NormalFloat 4-bit) quantization format

# QLoRA — Quantized LoRA



# QLoRA — Quantized LoRA




(a) QLoRA layer (b) QALoRA layer

# Applications of LoRA / QLoRA

- **Instruction tuning for resource-constrained settings:**
  - Enables fine-tuning large models on modest hardware (e.g., consumer GPUs, laptops).
  - Used for domain adaptation, personalization, and rapid prototyping.
- **Popular fine-tuned models:**
  - **Alpaca, Guanaco, Vicuna, Mistral:** All leverage LoRA/QLoRA for efficient instruction tuning.
- **Model merging and compositionality:**
  - Merge multiple LoRA adapters for multi-domain or multi-task capabilities.
  - Compose adapters for new tasks without retraining the base model.

# Evaluation Metrics

# Evaluation Metrics for LLMs

| | |
|---|---|
| **Perplexity** | Measures how well a model predicts text; lower is better |
| **BLEU** | Compares generated text to reference using n-gram overlap |
| **ROUGE** | Evaluates overlap with human-generated text; used for summarization |
| **METEOR** | Considers synonyms and word order; improved BLEU for translation |
| **BERTScore** | Calculates semantic similarity with contextual embeddings |
| **Human Evaluation** | Involves rating outouts for qualities like relevance and coherence |
| **Task-Specific Metrics** | E.g.. Exact Match for QA, Pass@k for code generation |
| **Prompt-Injected Evaluation** | Uses LLMs to assess outputs; scalable but may be biased |

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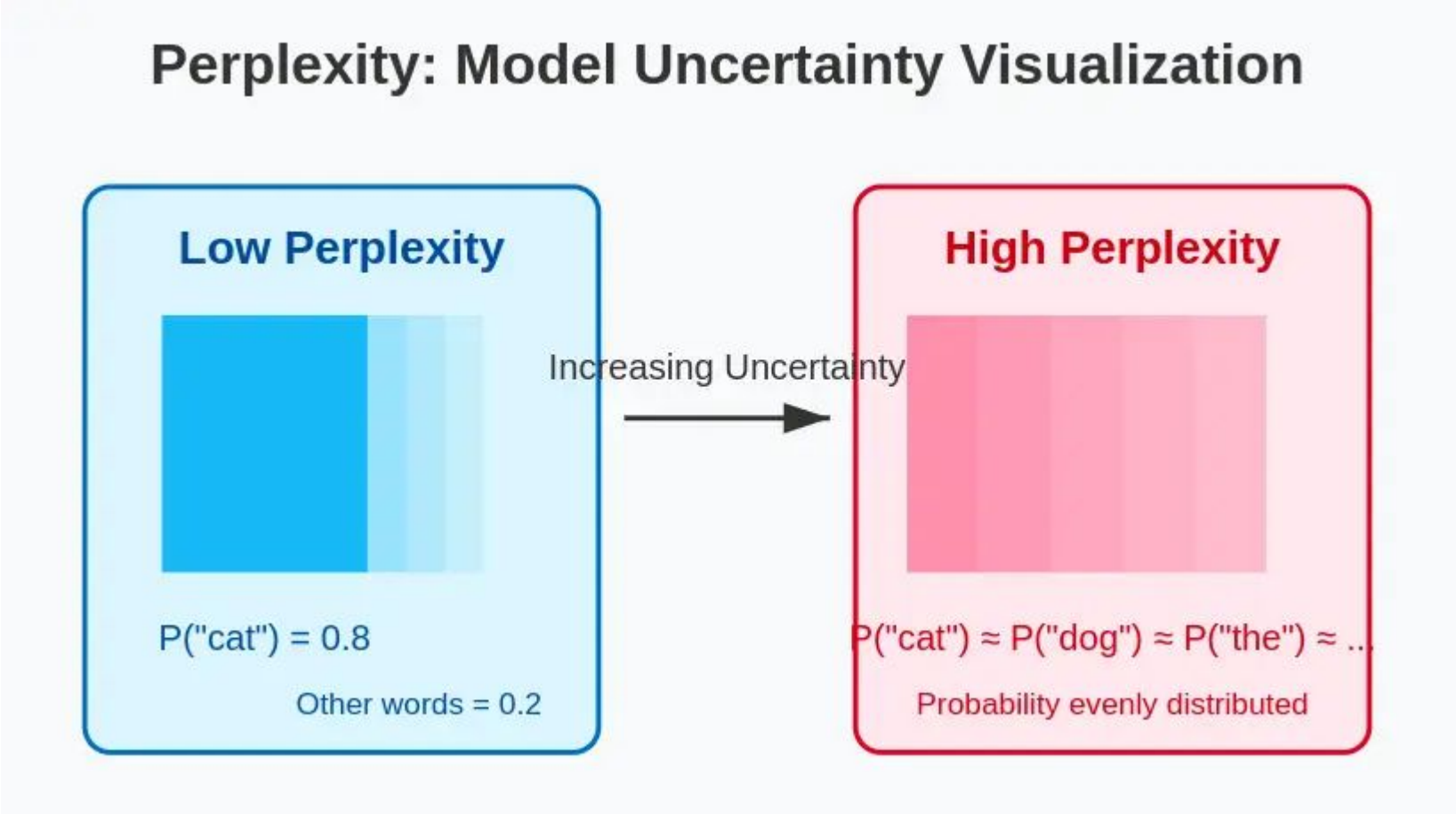
# Perplexity


Perplexity: Model Uncertainty Visualization
Low Perplexity
P("cat") = 0.8
Other words = 0.2
Increasing Uncertainty
High Perplexity
P("cat") ≈ P("dog") ≈ P("the") ≈ ...
Probability evenly distributed

# Perplexity

$$\text{perplexity} = \underbrace{\prod_{t=1}^{T} \left( \frac{1}{P_{\text{LM}}(\boldsymbol{x}^{(t+1)} \mid \boldsymbol{x}^{(t)}, \ldots, \boldsymbol{x}^{(1)})} \right)}_{\text{Inverse probability of corpus, according to Language Model}}^{1/T}$$

Normalized by number of words

# Perplexity

Dan Jurafsky

## Perplexity

The best language model is one that best predicts an unseen test set

- Gives the highest P(sentence)

Perplexity is the inverse probability of the test set, normalized by the number of words:

$$PP(W) = P(w_1 w_2 ... w_N)^{-\frac{1}{N}}$$

$$= \sqrt[N]{\frac{1}{P(w_1 w_2 ... w_N)}}$$

Chain rule:

$$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i | w_1 \ldots w_{i-1})}}$$

For bigrams:

$$\mathrm{PP}(W) = \sqrt[N]{\prod_{i=1}^{N} \frac{1}{P(w_i | w_{i-1})}}$$

**Minimizing perplexity is the same as maximizing probability**

# BLEU (Bilingual Evaluation Understudy)

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## BLEU

- N-gram overlap between machine translation output and reference translation
- Compute precision for n-grams of size 1 to 4
- Add brevity penalty (for too short translations)

$$\text{BLEU} = \min\left(1, \frac{\textit{output-length}}{\textit{reference-length}}\right) \left(\prod_{i=1}^{4} \textit{precision}_i\right)^{\frac{1}{4}}$$

- Typically computed over the entire corpus, not single sentences

# BLEU (Bilingual Evaluation Understudy)

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## BLEU Evaluation Metric

**Reference (Human) translation:**
The U.S. island of Guam is maintaining a high state of alert after the Guam airport and its offices both received an e-mail from someone calling himself the Saudi Arabian Osama bin Laden and threatening a biological/chemical attack against public places such as the airport .

**Machine translation:**
The American [?] international airport and its the office all receives one calls self the sand Arab rich business [?] and so on electronic mail , which sends out ; The threat will be able after public place and so on the airport to start the biochemistry attack , [?] highly alerts after the maintenance.

- Score is between 0 and 1 (sometimes normalized to a number between 0 and 100)
- **What percentage of MT output n-grams (text string clusters) can be found in the reference translation?**
- Usually calculated on ~**1000 test sentences**.
- Important to reward the right things and there is brevity penalty
- Getting larger word clusters to match provides better scores

Example from C. Wayne

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# LLM Evaluation - Metrics - ROUGE clipping

Reference (human):

It is cold outside.

Generated output:

cold cold cold cold

$$\text{ROUGE-1 Precision} = \frac{\text{unigram matches}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{1}{4} = 0.25$$

Generated output:

outside cold it is

$$\text{Modified precision} = \frac{\text{clip(unigram matches)}}{\text{unigrams in output}} = \frac{4}{4} = 1.0$$

# LLM Evaluation - Metrics - ROUGE-L

Reference (human):

It is cold outside.

Generated output:

It is very cold outside.

LCS:
Longest common subsequence

$$\text{ROUGE-L Recall:} = \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in reference}} = \frac{2}{4} = 0.5$$

$$\text{ROUGE-L Precision:} = \frac{\text{LCS(Gen, Ref)}}{\text{unigrams in output}} = \frac{2}{5} = 0.4$$

$$\text{ROUGE-L F1:} = 2\,\frac{\text{precision x recall}}{\text{precision + recall}} = 2\,\frac{0.2}{0.9} = 0.44$$

# ROUGE

ROUGE-N

$$= \frac{\sum_{S \in \{ReferemceSummaries\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count(gram_n)} \quad (1)$$

# References

These slides have been adapted from

- Bhiksha Raj & Rita Singh, 11-785 Introduction to Deep Learning, CMU

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## Credits

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